



Advancements and Prospects of Machine Learning in Medical Diagnostics: Unveiling the Future of Diagnostic Precision

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Abstract

Machine learning (ML) has emerged as a versatile and powerful tool in various fields of medicine, revolutionizing early disease diagnosis, particularly in cases where traditional diagnostic approaches face challenges due to unclear or overlapping symptoms. This survey provides a comprehensive overview of the wide-ranging applications of ML techniques in detecting and diagnosing various diseases at an early stage, highlighting their potential to transform healthcare practices. The survey commences with a comprehensive review of commonly used ML algorithms, emphasizing their relevance and adaptability in medical domains. With a focus on disease diagnosis, we delve into the specific implementation of ML algorithms for early detection in prominent diseases, including cancer, COVID-19, diabetes, kidney diseases, and heart diseases. By analyzing the current state of research and developments, this survey provides valuable insights into how ML algorithms are being employed to enhance disease diagnosis accuracy and efficacy. In the domain of cancer diagnosis, ML techniques have made significant strides in analyzing medical imaging data, genomic profiling, and predictive modeling. These advancements have led to improved cancer detection rates, enabling timely interventions and personalized treatment plans. Additionally, the survey explores the pivotal role of ML in addressing the challenges posed by the COVID-19 pandemic. ML-based automated screening tools have demonstrated efficiency in detecting potential cases, while predictive modeling has been instrumental in estimating disease progression and optimizing resource allocation. Furthermore, ML's contributions extend to chronic diseases such as diabetes, kidney diseases, and heart diseases, where it has shown promising results in predicting disease progression, enabling early intervention, and enhancing management strategies. In conclusion, this comprehensive survey showcases the transformative potential of ML in early disease diagnosis across various medical conditions. By providing valuable references and insights into future trends, it serves as a guiding resource for researchers and clinicians interested in leveraging ML technologies to improve patient care and make significant advancements in the field of medical diagnostics. With the capacity to decipher complex patterns and facilitate intelligent predictions, ML has emerged as a pivotal ally in the journey towards early disease detection and improved healthcare outcomes.

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1 Introduction

In the field of medicine, artificial intelligence (AI) mainly focuses on the development of methods for determining the correct behavior of the system when making a diagnosis [1]. A medical diagnosis is the identification of the disease or conditions that cause a person's symptoms and signs. Usually, the patient's medical history and physical examination provide diagnostic information. Diagnosing a medical condition can be difficult because many symptoms are ambiguous and can only be correctly identified by trained healthcare professionals. This can be a major problem in developing countries where doctors and other health professionals are scarce. Many people in these countries cannot get proper medical diagnosis and treatment for lack of healthcare professionals [2]. AI could be helpful for undeveloped regions [3]. The AI-based system [4, 5] is able to detect various diseases, such as kidney stone detection [6], heart disease prediction and detection [7], skin and lung cancer detection [8, 9], diabetes prediction and detection [10]. Recently, AI-based systems have played an important role in the correct diagnosis of COVID-19 [11, 12].

A branch of artificial intelligence called "ML" focuses on creating algorithms that allow computers to learn from data and make predictions. It is based on the idea that systems can automatically get better at a task over time by gaining experience and processing lots of data. Recent advances in ML technology have enabled computers to process and understand information on a par with humans. In some cases, these systems can recognize patterns and extract information with greater accuracy than trained experts. This is a significant step forward, as it demonstrates the ability of machines to process and understand complex information in new and novel ways. ML has become a powerful tool due to improvements in classical statistical modeling techniques, large amounts of data, growth in computing power, and advanced algorithm design [13]. ML algorithms can be trained on large datasets of medical images such as radiographs and MRI scans [14], to identify signs of disease or abnormalities. There are two types of problems in ML, classification and regression. Classification is predicting an outcome from a set of data, while regression is showing the relationship between dependent and independent variables.

While ML algorithms have been proven to be valuable tools in medical diagnosis [15, 16], it is essential to recognize that they should not replace human expertise and judgment. ML algorithms rely on the data they have been trained on and may not consider all relevant factors in the diagnostic process. Additionally, these algorithms are not infallible and can make errors. Therefore, it is crucial for healthcare professionals to exercise caution, review, and verify the results generated by ML systems to ensure accurate and reliable

diagnoses. The integration of ML into healthcare can be considered as a collaborative approach that combines the strengths of both algorithms and human expertise to achieve the best possible outcomes for patients.

1.1 Motivation

The main purpose of this thorough review is to provide a helpful guide for upcoming researchers, especially those who want to explore the unused possibilities of using ML for detecting diseases at an early stage. By delving deep into the application of ML algorithms across a diverse spectrum of diseases, this review endeavors to provide critical insights into the most effective strategies for early detection, thereby propelling the realm of medical diagnostics into a new era of precision and proactive care. In this pursuit, the review will address pivotal questions that hold paramount importance for researchers venturing into the realm of ML-powered early disease diagnosis. It is poised to be a beacon, illuminating crucial pathways for researchers seeking to incorporate ML into their diagnostic arsenal. To achieve this, the review will embark on a threefold exploration.

Firstly, it will chart a comprehensive course through the labyrinth of ML-based methods, guiding researchers in the art of applying these algorithms to diagnose a diverse array of diseases. Through a meticulous analysis of existing literature and studies, this section will equip future researchers with the sagacity to harness ML algorithms effectively within an expansive range of medical contexts.

Secondly, the review will cast a focused spotlight on five specific diseases: cancer, COVID-19, kidney disease, heart disease, and diabetes. Each of these diseases has been chosen with utmost consideration, reflecting their profound impact on humanity. Cancer, an enigmatic adversary with countless faces, strikes fear into hearts and calls for innovative diagnostic solutions. The unprecedented global turmoil wrought by COVID-19 emphasized the urgent need for rapid and accurate diagnostic tools. The stealthy yet devastating kidney disease, the relentless assault of heart disease, and the silent metabolic imbalance of diabetes underline the critical necessity of early detection. By delving into these human dangers, the review seeks to underscore the rationale for their selection and the urgency of pioneering ML-based diagnostic interventions.

Thirdly, the review will delve into the realm of ML algorithms that are being actively employed in the realm of smart medical applications for disease diagnosis. By deciphering the algorithms that have not only demonstrated success in clinical settings but have also been battle-tested in real-world medical scenarios, the review aims to facilitate future researchers in selecting the most appropriate tools for their own diagnostic pursuits. A pivotal focus within this review

lies in the rigorous evaluation of algorithmic performance in the context of medical diagnosis. The effectiveness of ML algorithms hinges on their reliability and accuracy, particularly when applied to the complex landscape of healthcare. Consequently, the review will meticulously examine the methodologies and metrics deployed in prior studies to gauge the sufficiency of these evaluation paradigms. The ultimate synthesis of this multifaceted exploration forms a robust foundation for empowering future researchers. By combining an exhaustive investigation of ML applications across a diverse array of diseases, insights into triumphant algorithms in real-world medical scenarios, and a comprehensive analysis of evaluation methodologies, this comprehensive review aims to arm researchers with the knowledge and tools to make judicious and informed decisions. This, in turn, will foster their contribution to the advancement of early disease diagnosis through the transformative power of ML.

1.2 In summary, the motivation behind this comprehensive review is threefold

- **Guiding Illumination:** It strives to serve as a guiding light for future researchers, offering them invaluable insights and answers to pressing questions pertaining to the application of ML-based methods in early disease diagnosis.
- **Disease-Specific Focus:** By spotlighting the distinct perils of cancer, COVID-19, kidney disease, heart disease, and diabetes, the review acknowledges the unique challenges posed by each disease and underscores the urgent need for innovative diagnostic approaches.
- **Pioneering Advancements:** Through its meticulous dissection of effective methodologies, disease-specific applications, successful ML algorithms, and robust evaluation paradigms, the review aims to drive the evolution of medical diagnostics and catalyze the development of dependable and accurate ML-based diagnostic systems.

1.3 The main motivation of our article can be summarized as

- Explain how ML can navigate the intricate world of medical diagnosis, showing how it can be used effectively.
- Bring attention to significant health concerns like cancer, COVID-19, kidney disease, heart disease, and diabetes. These diseases pose serious risks, and using ML for early diagnosis could make a real difference.

- Provide researchers with the tools to choose the best ML methods, ensuring their accuracy and reliability in real-world scenarios.
- Drive the progress of medical diagnostics by offering insights that will enable future researchers to contribute to the development of dependable and precise ML-based diagnostic systems.

In this review, we explain different ML algorithms used to diagnose different diseases. The paper is organized as follows. Section 2 gives the details of ML and DL algorithms. Section 3 gives details of publications by journal and year. Section 4 discusses the diagnosis of different diseases using ML algorithms. Section 5 presents the details of the publicly available datasets. Section 6 presents research challenges and future perspectives. Section 6 discusses the conclusions of this review.

2 Background

2.1 Types of ML

Diverse categories of ML models exist, encompassing supervised, unsupervised, semi-supervised, and reinforcement learning approaches.

Supervised Learning In this approach, the model is trained on labeled data and makes predictions based on that data. The objective is to establish a mapping function between input and output variables. Supervised learning is commonly used for tasks like image classification, where the model learns from labeled examples to classify new images accurately.

Unsupervised Learning This category of learning entails training the model using unlabeled data, allowing it to uncover concealed patterns and structures autonomously, devoid of human intervention. Unsupervised learning excels at tasks such as image recognition and clustering, where the model identifies similarities and differences within the data without any prior knowledge of the classes or labels.

Semi-Supervised Learning In this context, the model undergoes training through a blend of both labeled and unlabeled data sources. The labeled data is used to make predictions, while the unlabeled data aids in enhancing the model's accuracy and performance. The objective is to harness the wealth of unlabeled data in order to enhance the model's

overall performance, particularly in situations where acquiring labeled data proves expensive or time-intensive.

Reinforcement Learning In the realm of reinforcement learning, an agent acquires knowledge through iterative trial and error engagements with its surroundings. By taking actions and receiving subsequent rewards or penalties contingent on its actions, the agent adapts its decision-making approach to optimize the cumulative reward signal. Reinforcement learning finds applications in various domains, such as robotics, gaming AI, and recommendation systems.

Figure 1 illustrates the different methods of ML algorithms, depicting the relationships and distinctions among supervised, unsupervised, semi-supervised, and reinforcement learning approaches.

2.1.1 Support Vector Machine (SVM)

SVM is a ML algorithm used for classification tasks. SVM was introduced in the 20th century by Vanpik [17]. The primary objective of SVM is to identify an optimal hyperplane that effectively divides the gathered data into distinct classes. The space between this hyperplane and the nearest data points from each class is referred to as the support vector. The selection of the hyperplane aims to maximize the margin, denoting the gap between the vector and the hyperplane itself. SVM handles nonlinearly separable data.

It transforms the data into a high-dimensional space where linear boundaries can be found.

2.1.2 Decision Tree (DT)

A DT is a supervised learning model employed to address both classification and regression tasks. DT follows the divide-and-conquer strategy [18]. DT works by dividing the source data into subsets based on the most significant attributes. This splitting is done in a recursive manner, hence it is called recursive partitioning. Each internal node is characterized as a dataset, with branches representing the decision made and leaves representing the final result or prediction. Decision Trees are popular for modeling and visualizing complex decision-making processes because they are easily interpretable and can handle both categorical and numerical data.

2.1.3 Naïve Bayes

Naïve Bayes [19] is a probabilistic ML algorithm that is based on the Bayes theorem. This theorem states that which states that the probability of a hypothesis (e.g., a class label in classification) given observed evidence (e.g., features of an instance) is proportional to the prior probability of the hypothesis and the likelihood of the evidence given the hypothesis.

The NB algorithm is used to predict the class label of an instance based on its feature. This model makes a strong

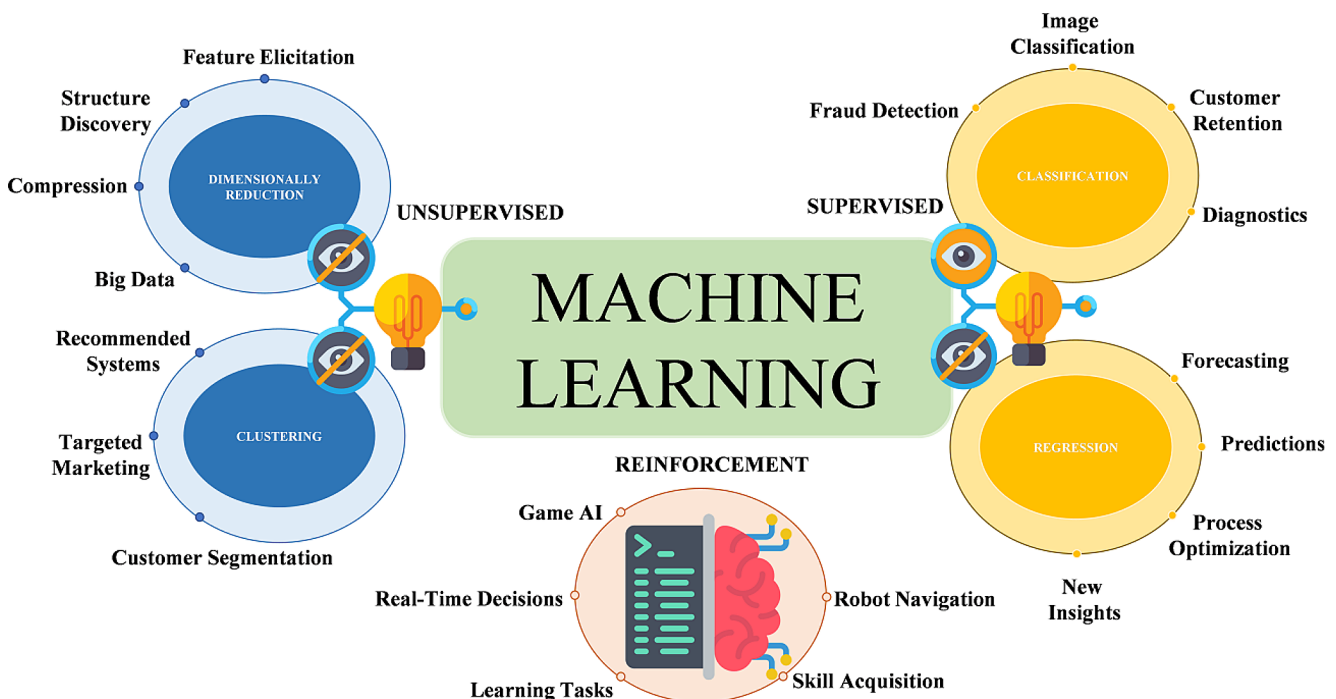


Fig. 1 Different methods of ML algorithms

assumption of independence between features, which is why it is called naive. This independence assumption doesn't hold in reality. This algorithm works well when there is high dimensionality and a large amount of data to classify.

2.1.4 AdaBoost

AdaBoost, short for Adaptive Boosting, stands as a widely recognized ensemble learning algorithm in ML. It orchestrates the amalgamation of multiple weak models, culminating in the construction of a robust model that contributes to precise predictions. This algorithm gives more weight to the samples that are harder to classify so that the model focuses more on those samples. This process is repeated several times and the weights are updated at each iteration until a satisfactory model is obtained. The final model is a weighted average of the individual weak models [20].

2.1.5 K-nearest neighbor

KNN is an ML classification algorithm that assigns a label to a data point based on the majority vote of the K nearest neighbors used for prediction. The algorithm identifies K data points within the training set that exhibit the closest proximity to the new data point. Subsequently, it leverages the class labels of these K neighbors to formulate a prediction. This prediction is determined through a majority vote among the classes of these K neighbors. It is important to choose an appropriate value of K because K has a significant impact on the performance of the algorithm [21].

2.1.6 Logistic Regression

Logistic Regression (LR) serves as a statistical model within ML, elucidating the association between a continuous dependent variable and independent variables. The independent variables, often referred to as predictors or features, play the role of prognosticators in anticipating the outcome. LR is used for 2-class classification. It first calculates the linear combination between the independent variables and the coefficient, known as the logit. It then predicts the outcome by transforming the output through a logistic/sigmoid function [22].

2.2 Deep Learning Overview

Deep learning (DL) represents a remarkable branch of AI and ML that has garnered significant attention due to its ability to mimic the structure and function of the human brain [23]. It has been demonstrated extraordinary potential in various applications, particularly in fields such as medical disease prediction, computer vision, and decision-making.

DL algorithms, built on the concept of Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN), have proven to be highly effective at learning complex relationships within data and making accurate predictions. The foundation of DL lies in neural networks, which consist of interconnected processing nodes arranged in layers. Each node, or neuron, processes input data and passes the result to subsequent layers, eventually leading to the generation of an output. These networks excel at learning intricate patterns from raw data, eliminating the need for manual feature engineering, a process that traditionally required domain expertise and considerable effort [24]. The process of training DL networks is central to their success. It involves presenting the network with vast amounts of labeled data and adjusting its internal parameters, namely the weights and biases, to minimize prediction errors. This optimization process is typically achieved through gradient descent and backpropagation algorithms, which iteratively fine-tune the model's parameters until it becomes capable of accurate predictions.

One of the most significant breakthroughs facilitated by DL is in the field of medical diagnosis, particularly in medical image processing. By leveraging DL models, researchers and healthcare professionals have made significant strides in diagnosing various medical conditions using different medical imaging modalities, such as Magnetic Resonance Imaging (MRI) and chest radiographs [25–28]. These models have showcased unprecedented accuracy, often outperforming human experts. Despite these accomplishments, DL models in medical diagnosis are still in their early stages and have their limitations. While they excel at accuracy, their interpretability remains a challenge. Understanding the decision-making process of complex neural networks can be elusive, leading to concerns about the transparency and trustworthiness of their results. As such, extensive testing and validation are necessary before these models can be widely integrated into clinical practice. It is essential to interpret their outputs in conjunction with a patient's medical history and other diagnostic information to ensure precise and reliable diagnoses.

As illustrated in Fig. 2, the development of ML and DL has evolved over time, with DL emerging as a powerful extension of ML. The timeline depicts the progress made in both fields, showcasing how DL has transformed various domains and opened up new possibilities for researchers, developers, and practitioners alike. In conclusion, DL stands as a revolutionary advancement in AI and ML. Through its utilization of neural networks and the ability to automatically extract complex patterns from data, DL has enabled breakthroughs in numerous AI applications. In the medical field, DL has shown promise in diagnosing diseases through medical image processing, revolutionizing healthcare practices. However, challenges related to model interpretability

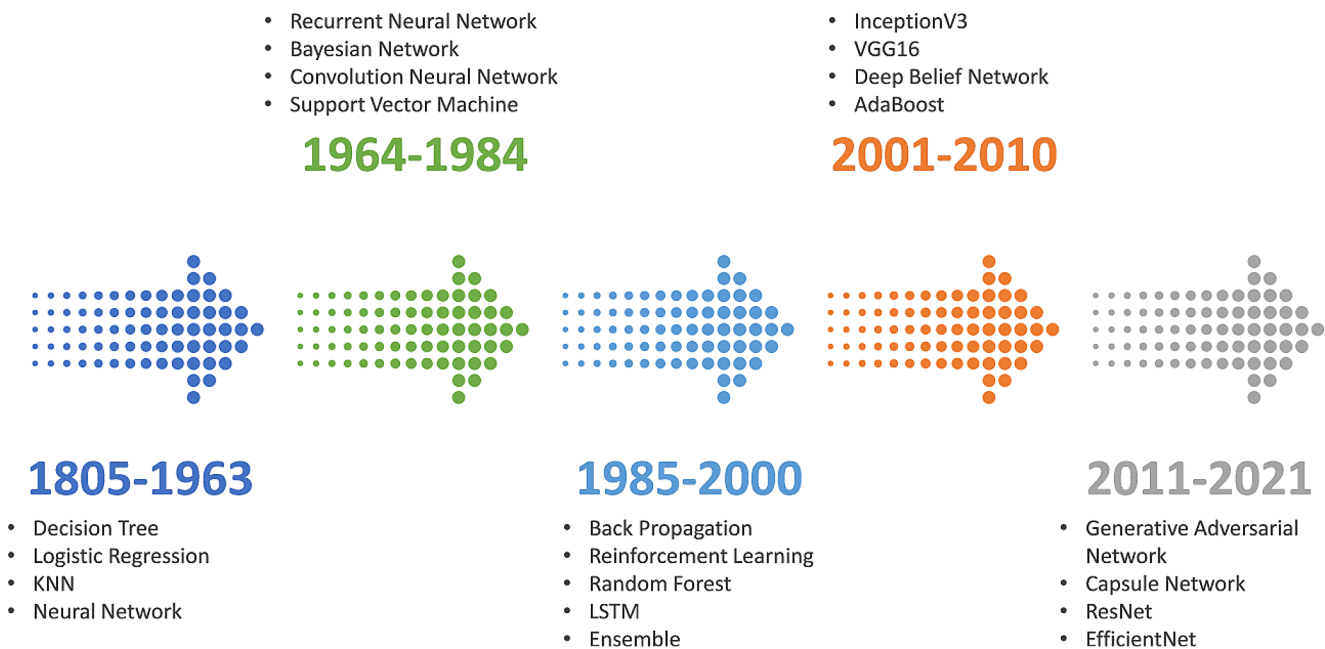


Fig. 2 A diagram of the development timeline for ML and DL models

and rigorous validation remain critical areas of focus. As researchers continue to explore and refine DL techniques, its impact on AI and diverse industries will undoubtedly continue to be profound, fueling innovation and driving future advancements.

2.2.1 Convolutional Neural Networks

CNNs represent a category of powerful DL models, mainly employed for tasks in computer vision. Their applications encompass a wide range of tasks, including image classification and segmentation [29]. They are specifically designed to process visual data efficiently by exploiting the spatial correlations present in images [30]. A typical CNN consists of multiple layers. The first layer is the input layer, which takes the raw image data as input. Then, a series of convolutional layers follow. In each convolutional layer, the model utilizes a collection of tunable filters, also referred to as kernels, to process the input image, producing feature maps that highlight different patterns or features [31]. Pooling layers are frequently interspersed between convolutional layers to perform downsampling on the feature maps, effectively reducing their spatial dimensions while preserving critical information. This downsampling process not only alleviates the computational load but also enhances the network's resilience to input variations, making it more robust in handling diverse visual patterns [32]. The outcomes obtained from these layers are commonly flattened and then fed into one or multiple fully connected layers. These fully

connected layers process the high-level features and make the final predictions, depending on the specific task.

To introduce non-linearity, activation functions are applied after each layer's output. The widely adopted activation function in CNNs is the Rectified Linear Unit (ReLU) due to its straightforwardness and its ability to address the vanishing gradient issue effectively. By setting all negative values to zero, ReLU allows the network to efficiently learn intricate representations [33, 34].

CNNs are typically trained using various loss functions, depending on the task. For image classification, the softmax function combined with the categorical cross-entropy loss is commonly used to measure the dissimilarity between predicted and true class probabilities.

In the domain of regression tasks, it is commonplace to utilize loss functions such as Mean Squared Error (MSE) or Mean Absolute Error (MAE) [35]. Training CNNs involves forward and backward passes through the network using an optimization algorithm like Stochastic Gradient Descent (SGD) or its variants. The goal is to minimize the chosen loss function and update the network's parameters to improve its performance on the task.

In summary, CNNs are a powerful class of DL models for computer vision tasks. They consist of convolutional layers to extract relevant features, pooling layers for downsampling, fully connected layers for decision-making, and ReLU activations to introduce non-linearity. The choice of loss functions depends on the specific task, and CNNs are trained to minimize these losses through optimization algorithms. Overall, CNNs have revolutionized the field of

computer vision and continue to play a vital role in various real-world applications [36]. Figure 3 illustrates the basic CNN architecture.

2.2.2 Transfer Learning

Transfer learning constitutes a ML approach wherein insights gained from training a model on a particular task or domain are harnessed to elevate performance on a separate yet interconnected task or domain [37]. Transfer learning offers a significant advantage by reducing the necessity for a large amount of labeled data to achieve satisfactory performance on the target task [38, 39]. Pre-training on a large and diverse dataset allows the model to learn general features that are transferable to a wide range of tasks. As a result, when the model is fine-tuned on the target task with a smaller dataset, it can adapt its learned representations to the specific nuances of the new data, leading to improved performance. Transfer learning also enables the application of DL to tasks with limited data availability. Training deep neural networks from scratch typically requires a large dataset to prevent overfitting, but in many real-world scenarios, collecting such vast amounts of data might be challenging or expensive. By utilizing pre-trained models, which have already learned useful features, transfer learning allows us to tackle tasks with limited labeled data effectively.

Another advantage of transfer learning is that it can help mitigate the problem of vanishing gradients. Deep neural networks often have numerous layers, and during training, gradients can become very small, making it difficult for the model to learn effectively. Pre-training on a related task helps in initializing the model's weights in a way that provides a better starting point for the optimization process during fine-tuning, making training more stable and efficient. Furthermore, transfer learning allows the application of knowledge learned from one domain to another, leading to better generalization. For example, features learned from natural image datasets can be transferred to medical imaging tasks or satellite imagery analysis, even though the source and target domains are quite different. This transfer

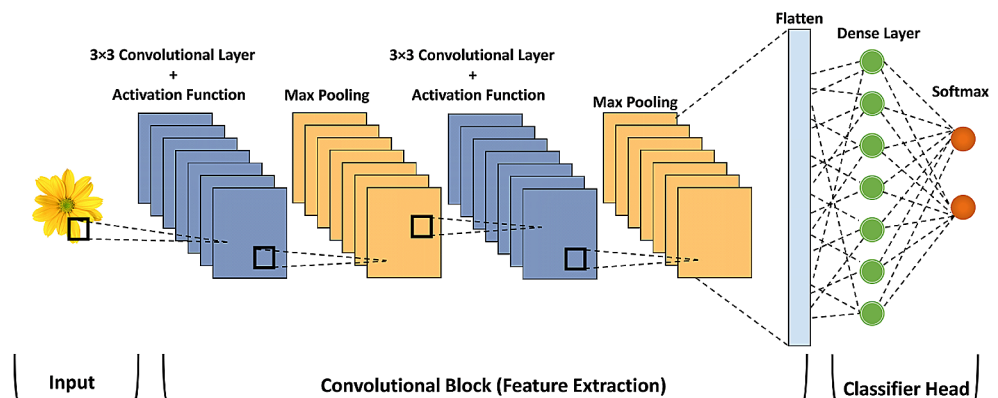
of knowledge helps in building more robust and versatile models that perform well in various scenarios [40, 41]. Its advantages include the reduction of data requirements, the ability to handle limited labeled data, better initialization for training, and improved generalization across different domains, making it a crucial tool for advancing artificial intelligence applications.

2.2.3 Emerging Trends in CNNs

Pre-trained models have emerged as a game-changer in the field of AI, especially for computer vision tasks. The idea behind these models is to pre-train a neural network on a large and diverse dataset, such as ImageNet, which contains millions of images belonging to thousands of different categories. During pre-training, the model learns to extract and encode meaningful features from the data, effectively capturing a general understanding of visual patterns and structures. The advantages of pre-training are manifold. Firstly, it significantly reduces the need for massive labeled datasets when tackling new tasks. Moreover, pre-trained models act as knowledge repositories that encapsulate valuable information about the low-level and high-level features found in the training data. As a result, they offer a head start for new tasks, especially in domains where gathering labeled data can be expensive, time-consuming, or simply impractical. Additionally, pre-trained models foster better generalization. The features learned during pre-training tend to be more robust and transferable across different datasets and domains. This transfer of knowledge leads to models that exhibit improved performance, particularly when the target task's dataset is small or dissimilar from the original training data.

These pre-trained architectures, such as AlexNet, VGG, InceptionV3, ResNet, DenseNet, InceptionResNetV2, Xception, and MobileNet, have become standard tools in the DL toolkit. Researchers and practitioners commonly use them as the starting point for their specific computer vision projects, saving substantial computational resources and time that would otherwise be spent on training from scratch.

Fig. 3 CNN architecture



- AlexNet: Introduced in 2012 [38], AlexNet is one of the pioneering deep CNN. It plays a significant role in popularizing DL for computer vision tasks. AlexNet is characterized by its architecture, which comprises five convolutional layers, all using ReLU activations. This arrangement results in a deeper network compared to earlier architectures, contributing to its ability to learn more intricate features and patterns from input data.
- VGG (Visual Geometry Group): VGG [42] is a deep architecture proposed by the Visual Geometry Group. It is known for its simplicity and uniformity, consisting of multiple layers with small 3×3 convolutional filters and max-pooling layers. VGG variants with different depths, such as VGG16 and VGG19, are widely used.
- InceptionV3: Also known as GoogLeNet, InceptionV3 [43] was introduced by Google researchers. It features the Inception module, which uses multiple filter sizes within the same layer to capture different-scale features. This architecture is efficient and has excellent performance on various tasks.
- ResNet (Residual Network): ResNet [44] is famous for introducing residual connections, which allow the model to skip some layers during training. This innovation addresses the vanishing gradient problem and made it possible to train much deeper networks. ResNet50 and ResNet101 are common variants.
- DenseNet: DenseNet [45] is designed to maximize feature reuse by connecting each layer to every subsequent layer in a feed-forward fashion. This densely connected architecture fosters better gradient flow, reduces the network complexity, and encourages feature propagation throughout the network.
- InceptionResNetV2: Combining ideas from both Inception and ResNet, InceptionResNetV2 [46] achieves strong performance while being computationally efficient. It incorporates residual connections and inception modules, offering a potent combination of features.
- Xception (Extreme Inception): Xception represents a development of the Inception architecture and introduces depthwise separable convolutions [47]. These separable convolutions in Xception are designed to dramatically decrease the computational cost while still preserving high accuracy in the model's predictions.
- MobileNet: MobileNet is designed for mobile and embedded vision applications, emphasizing efficiency and small model size [48, 49]. It employs separable convolutions to reduce the computation complexity, making it suitable for resource-constrained environments.

In summary, pre-trained models have revolutionized the landscape of DL, specifically in the realm of computer vision. Their ability to transfer knowledge from extensive

pre-training to new tasks with limited data has made them indispensable tools for building high-performing and resource-efficient models in various real-world applications. As DL continues to advance, pre-trained models will undoubtedly remain a crucial component in accelerating progress and making cutting-edge AI accessible to a broader audience.

2.3 Model Evaluation Metrics

In this section, we explain the performance metrics of the models. Model evaluation metrics are mathematical functions used to evaluate the performance of ML models. Accuracy, precision, sensitivity, and f1 score are the most commonly used evaluation metrics. The accuracy rate is the ratio between the correctly identified samples and the total samples. Precision is defined as the ratio of true positive predictions to all positive samples. The sensitivity is used to predict the actual positive samples for each available sample. The F1 score is used to evaluate the ability of the model by combining precision and recall. FP false positives, TN true negatives, FN false negatives, and TP true positives were used to calculate the performance metrics of the model. All metrics can be calculated as follows:

$$Accuracy = \frac{TN + TP}{TN + TP + FN + FP} \quad (1)$$

$$Precision (PPV) = \frac{TP}{TP + FP} \quad (2)$$

$$Specificity = \frac{TN}{TN + FP} \quad (3)$$

$$Sensitivity (Recall) = \frac{TP}{TP + FN} \quad (4)$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (5)$$

$$NegativePredictiveValue (NPV) = \frac{TN}{TN + FN} \quad (6)$$

$$MCC = \frac{(TP \times TN - FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (7)$$

3 Bibliometric Analysis

The Web of science and Scopus databases were used to collect references for statistical analysis. We studied the published papers using subject area, year and journal.

3.1 Subject Area

While ML has gained significant traction in healthcare applications, there are still several areas of ML-based research that require further exploration. Figure 4 provides an overview of the ML algorithms utilized across various fields. Interestingly, the dominant fields that heavily emphasize ML methods are computer science (40%) and engineering (31.2%). This indicates the widespread adoption and focus on ML techniques in these disciplines, showcasing the potential for interdisciplinary collaboration and knowledge transfer to further advance ML-based research in healthcare and other domains.

3.2 Publications by Year

Figure 5 provides an insightful representation of the exponential growth in published papers utilizing ML methods for disease diagnosis since 2014. The data collection process involved searching through web of science and Scopus databases using keywords such as “ML in medical applications” and “ML-based disease diagnosis.” The graph effectively demonstrates the increasing interest and research activity

in this field, highlighting the expanding body of knowledge and the significance of ML in medical research.

3.3 Publications by Journals

The data collection process involved gathering papers from the most popular and reputable journals published over the span of the past eight years. The analysis revealed that PLOS ONE, Nature Scientific Reports, and IEEE Access emerged as the most productive journals in terms of the number of papers published on ML-based disease diagnosis. Figure 6 visually presents the distribution of published papers across various journals. The graph provides a quantitative comparison, allowing researchers to identify the leading journals in the field and gauge the level of activity and interest within each publication. The prominence of Expert systems with applications, multimedia tools and applications, PLOS ONE, Nature Scientific Reports, Artificial intelligence in medicine and IEEE Access can be attributed to several factors, including their wide readership, interdisciplinary focus, and commitment to open access publishing. These journals likely attract a diverse range of researchers from different domains, contributing to the growing body of literature on

Fig. 4 ML applied in different fields

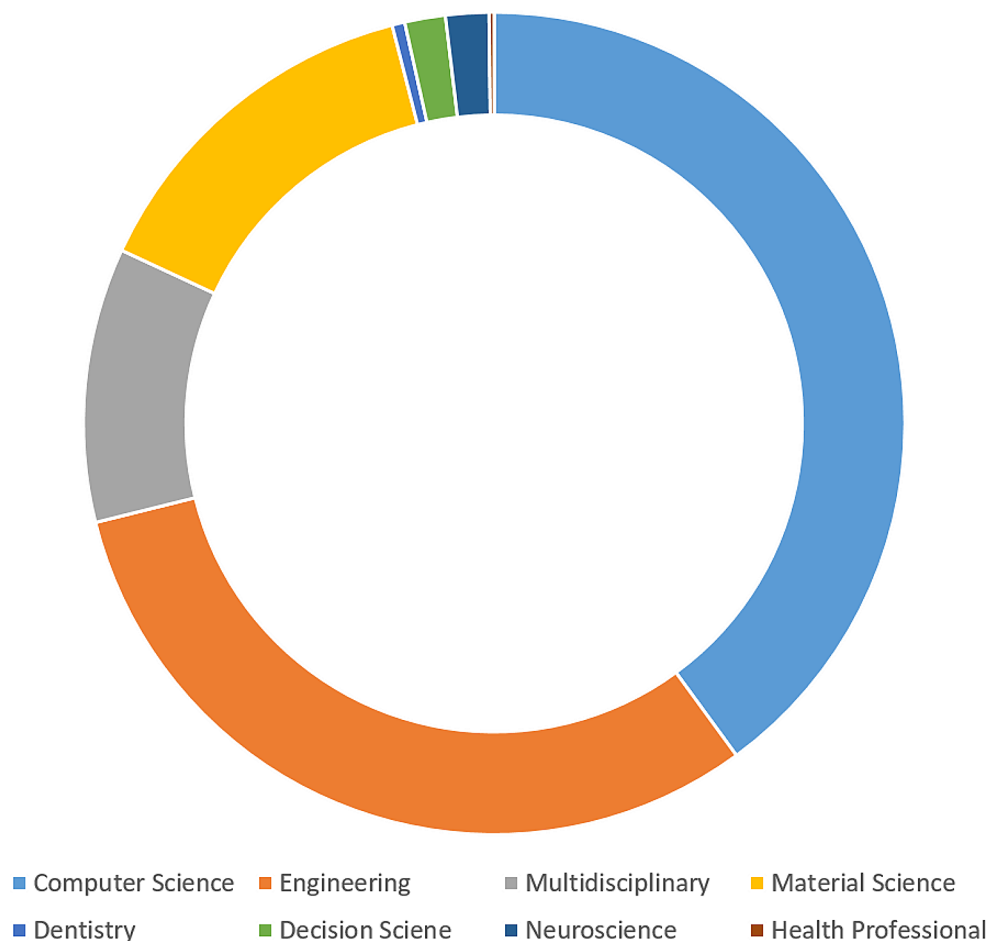


Fig. 5 ML papers published between 2014 and 2022

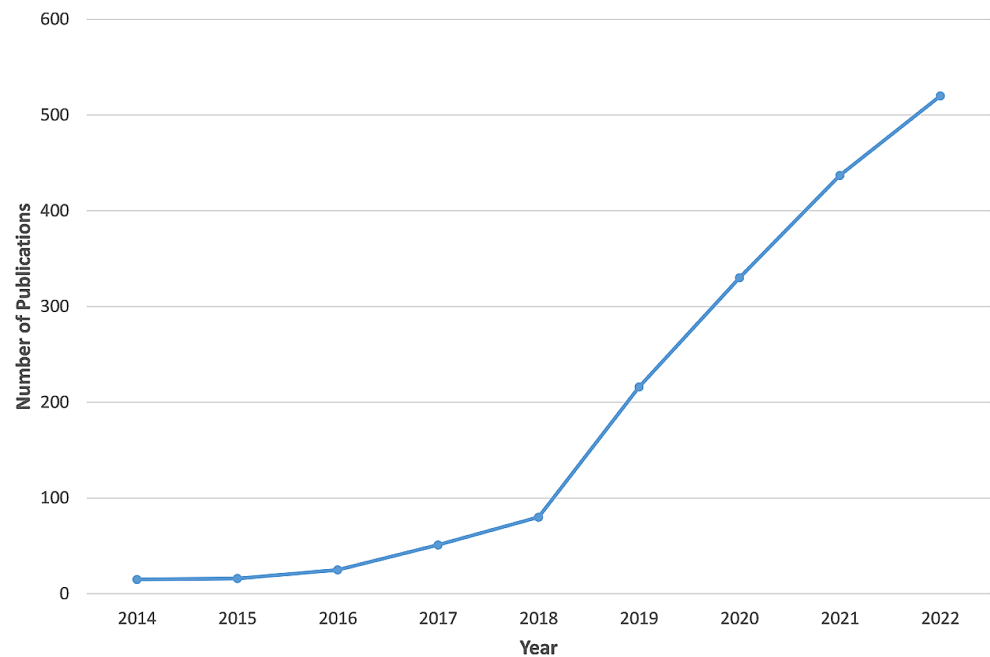
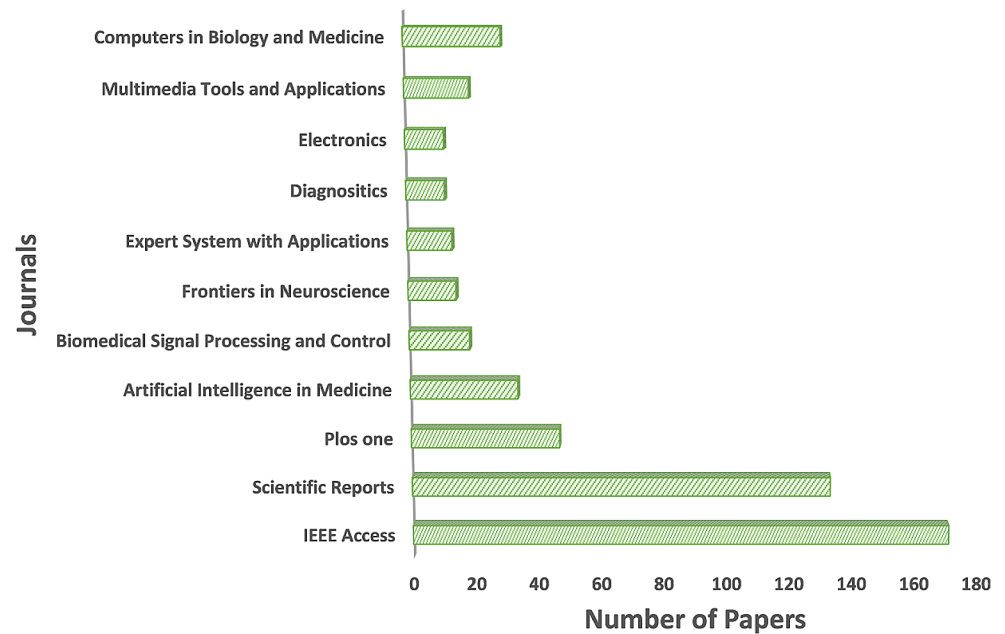


Fig. 6 Publications in different journals



ML applications in disease diagnosis. The significance of this finding lies in the recognition of these key journals as valuable resources for researchers interested in ML-based disease diagnosis. Researchers can target these journals for publication, ensuring their work reaches a broad audience and contributes to the collective knowledge in the field. Overall, Fig. 5 serves as a valuable visual representation of the publication trends in ML-based disease diagnosis, shedding light on the leading journals that have contributed to the dissemination of research findings in this rapidly evolving domain.

4 Exploring the Potential of ML in Healthcare Diagnostics

4.1 Cancer

Islam et al. [50] conducted a comparative analysis of five ML models using the Wisconsin Breast Cancer dataset (WBCD) sourced from the UCI ML Database. The findings demonstrate that the ANN achieved the utmost accuracy, reaching 98.57%. Sakri et al. [51] conducted a study to assess the precision of diverse data mining algorithms for forecasting breast cancer recurrence. In this endeavor, a

feature selection technique known as particle swarm optimization (PSO) was integrated into three widely recognized ML classifiers, enhancing the overall accuracy. The goal was to increase the accuracy of the prediction model. With the implementation of PSO, the authors identified four features that were best for the classification task. With PSO applied, the accuracy results for NB, Fast decision tree learner, and K-NNs improved to 81.3%, 80%, and 75%, respectively. Yue et al. [52] assessed a range of ML techniques for their efficacy in predicting breast cancer through the utilization of the WBCD dataset. The investigation highlighted the deep belief networks approach coupled with a DBNs-ANN as the most precise method, yielding an exceptional accuracy of 99.68%. The SVM technique, combined with a two-step clustering algorithm, achieved a classification accuracy of 99.10%. The authors also reviewed an ensemble technique, which involved combining SVM, Naive Bayes, and J48 methods using a voting mechanism, resulting in an accuracy of 97.13%. Rashmi et al. [53] propose two methods to classify the stages of melanoma cancer. The first method categorizes melanoma into either stage 1 or 2, while the second method expands this to classify melanoma into one of three stages: 1, 2 or 3. The author introduces a new method using a CNN algorithm combined with the SMTP loss function. The study compares the results of the new method with other loss functions that are designed for classification problems and finds that the proposed method is more effective and efficient. Fateh et al. [54] used data on breast cancer tumors collected by Dr. William H. Walberg at the University of Wisconsin Hospital. Various ML techniques were applied to the data to predict breast tumor type. The LR model with all features included achieved the accuracy rate of 98.1%, indicating the potential for improved breast cancer detection. Khan et al. [55] proposed a framework that uses pre-trained CNN architectures to extract important features from images, which are then fed into a new classifier for the classification of cancer cells. It has been observed that it outperforms other DL models in terms of accuracy in detecting and classifying breast tumors in cytology images, based on experiments performed on standard benchmark data sets. Alamin et al. [56] introduced an ensemble model designed to effectively detect lung and colon cancers. The model underwent evaluation using the histopathological (LC25000) lung and colon dataset, yielding impressive accuracy rates: 99.05% for lung cancer, 100% for colon cancer, and 99.30% for both lung and colon cancers. Notably, the proposed model outperformed existing models, signifying its potential application in clinical settings for accurate cancer diagnosis. Bansal et al. [57] presented a DL model based on VGG19 and also used other feature extraction methods (SIFT, SURF, ORB, Shi-Tomasi). The extracted feature were classified using

different ML algorithms, and the results showed that RF with collaborative features had an accuracy of 93.73%, outperforming other classifiers. The findings suggest that a single feature extractor is insufficient and that a mixed method using both DL features and classical handmade features is preferable for picture classification. Ramdas et al. [58] introduced a CNN-LSTM model for the purpose of detecting brain tumors by analyzing 3264 MRI images of the brain. The dataset was partitioned into training and testing subsets. The empirical outcomes of the experimentation reveal that the proposed model exhibits superior accuracy compared to both existing CNN and RNN models. Hanan et al. [59] presents a computer-aided diagnosis method for breast cancer classification using a combination of DNN and TL on a publicly available dataset. The results shows that the proposed method achieved the best average accuracy for binary classification of benign or malignant cancer cases at 99.7% for ResNet. The average accuracy for multi-class classification were 97.81% for ResNet. George et al. [60] proposed and tested a new methodology that combines probabilistic neural networks and SVM for cell nuclei detection and breast cancer classification. The experiments conducted on breast cytological images yielded positive results, as the error rate, correct detection rate, sensitivity, and specificity were all significantly improved. The paper [61] thoroughly explores feature selection techniques and ML models for breast cancer detection. The methodology includes PCA, SVD, Chi2 feature selection, and SMOTE up-sampling to improve ML performance. The approach offers a reliable tool for timely breast cancer detection, aiding prompt treatment interventions. The research [62] investigates the impact of dimensionality reduction methods on cancer prediction using ML models. Evaluating classifiers with various metrics, the study demonstrates that autoencoder outperforms PCA and PCA with a kernel, positively influencing classifier performance in high-dimensional data scenarios, enhancing ML classification model results. The study proposes a computer model [63] for predicting TIIDM illness using ML and ensemble learning techniques, and are explored for TIIDM prediction, and synthetic minority oversampling and K-fold cross-validation are used to balance classes and validate results. The proposed methodology serves as an effective tool for early diabetes prediction and can be adapted to datasets with similar characteristics. The study [64] utilizes electronic health records to improve stroke prediction, identifying age, heart disease, average glucose level, and hypertension as key factors. The perceptron neural network with these attributes achieves superior accuracy and lower miss rate compared to benchmark algorithms, while dataset imbalance is addressed using sub-sampling techniques. The paper emphasizes reproducibility by providing accessible online simulation source code. The

paper introduces DFELMDA [65]. An innovative computational method is introduced in this study to predict miRNA-disease connections. The approach leverages deep forest ensemble learning combined with autoencoder techniques. DFELMDA employs two deep autoencoders for feature representation and utilizes Deep Random Forest to provide prediction scores, outperforming traditional approaches on the HMDD dataset. Case studies on various disease types further validate its exceptional capacity to predict illness-associated miRNA-disease relationships, making it a valuable tool in computational biology. The study [66] introduces a DL architecture for predicting 72-hour ED bounce-backs using structured and unstructured EHR data, aiding emergency care by early identification of high-risk patients. Integrating structured vitals and demographics data with unstructured physician notes enhances prediction accuracy. Focusing on ED readmissions within 72 h fills a research gap in the literature. In [67], a novel supervised feature selection technique based on manifold learning is proposed, aiming to maximize same-class sample proximity within the selected subset. It effectively reduces computational complexity in microarray datasets while improving diagnostic accuracy. Comparative evaluation against ten existing algorithms on seven microarray datasets confirms its superior performance, establishing its promise as an effective feature selection approach for microarray data analysis. This paper [68] presents an innovative dose distribution prediction network tailored for cervical cancer radiotherapy. Leveraging an auxiliary segmentation task, the network learns PTV and OAR contour and position information, enhancing dose prediction accuracy. The approach incorporates CtFF modules with an attention mechanism, effectively fusing shared encoder content, auxiliary segmentation contour and position, and main dose prediction information in the decoding section. Extensive testing on internal cervical cancer datasets and publicly available head and neck cancer datasets validates the approach's superiority over existing techniques. The paper [69] presents a two-phase progressive transfer strategy employing transfer learning (TL) to predict cervical cancer doses with limited data. It transfers knowledge from rectum cancer to the target domain, enhancing prediction accuracy. The proposed approach utilizes the WFTM for effective knowledge transfer between networks, demonstrating its usefulness in cervical cancer dose prediction through TL on 130 cancer patients. The paper introduces TransDose [70], a novel CT image-based dose distribution prediction method. It utilizes a super-pixel-based graph convolutional network (GCN) for category-specific feature extraction and incorporates the Transformer for improved long-range sequence modeling, enhancing inter-slice continuity. The approach demonstrates its generalizability and advantages on real datasets, providing an

end-to-end mask-free solution with an auxiliary anatomical information extraction branch for crucial location and boundary information to the primary dose prediction branch. The Table 1 showcases a range of methods employed for cancer prediction and classification, highlighting their respective advantages and achievements in accuracy and efficiency. This section of the review paper delves into the multifaceted landscape of cancer research, encompassing three pivotal subsections:

“Lung Cancer”: Within the context of ML, this subsection investigates how computational models are enhancing our understanding of genetic mutations, aiding early detection, and guiding personalized treatment strategies.

“Breast Cancer”: This subsection delves into the intersection of ML and breast cancer, showcasing how algorithms are unraveling the complexities of molecular subtypes, optimizing treatment selection, and facilitating predictive outcomes.

“Brain Cancer”: In the realm of ML, this section spotlights how data-driven approaches are transforming brain tumor diagnosis, leveraging precision medicine algorithms, and advancing neuroimaging techniques to reshape treatment paradigms.

These subsections collectively provide a comprehensive overview of the current state of research and innovation in the domains of lung, breast, and brain cancer, underscoring the dynamic nature of cancer research and its profound implications for patient care.

4.1.1 Breast Cancer

In [71], they proposed a novel ML based framework, diagnoses breast cancer using full-field digital x-ray breast images, leveraging the self-attention mechanism of ViT and ensemble transfer learning of CNNs to generate precise high-level deep features. The framework's prediction performance was validated on real breast cancer images, achieving high result for binary and multi-class using INbreast mammograms for evaluation. In [72], propose a hybrid DL CAD system for breast lesions, combining Residual Convolutional Network and Transformer Encoder with MLP to distinguish between binary and multiclassification breast cancer. Author evaluate AI model against pre-trained CNN using CBIS-DDSM and DDSM datasets, achieving high accuracy for binary and multiclass prediction. The study [73] proposes “feature Score computation,” a novel method to re-rank Google search engine results by calculating scores based on page characteristics, effectively elevating relevant products and increasing corpus relevancy from 88 to 99%. In [74], proposes a contemporary CAD architecture utilizing DL to assist radiologists in breast cancer detection. Four experiments are conducted, incorporating pre-trained Deep CNNs

Table 1 Advances in Cancer Prediction and Classification Methods

Reference Number	Method Name	Advantage
[50]	ANN (ML model)	Highest accuracy: 98.57%
[51]	PSO with ML classifiers	Improved accuracy with feature selection
[52]	DBNs-ANNs, SVM, Ensemble	High accuracy results
[53]	CNN with SMTP loss function	More effective and efficient classification
[54]	Various ML techniques	LR achieved accuracy of 98.1%
[55]	Pre-trained CNN with new classifier	Outperforms other DL models
[56]	Ensemble model for lung and colon cancer	High accuracy in cancer identification
[57]	DL model (VGG19) with ML algorithms	RF with collaborative features at 93.73%
[58]	CNN-LSTM for brain tumor detection	More accurate than existing models
[59]	DNN and TL for breast cancer classification	High average accuracy for binary and multi-class
[60]	Probabilistic NN and SVM for cancer classification	Improved detection and classification
[61]	Feature selection and ML models	Reliable tool for timely breast cancer detection
[62]	Autoencoder for cancer prediction	Autoencoder outperforms PCA
[63]	Ensemble learning for TIIDM prediction	Effective tool for early diabetes prediction
[64]	Perceptron NN for stroke prediction	Superior accuracy with key factors
[65]	Deep forest ensemble for miRNA-disease	Outperforms traditional approaches
[66]	DL architecture for ED bounce-backs	Early identification of high-risk patients
[67]	Feature selection based on manifold learning	Reduces complexity and improves accuracy
[68]	Dose distribution prediction for cancer therapy	Enhanced dose prediction accuracy
[69]	Progressive transfer strategy for dose prediction	Improved accuracy with limited data

in the first, using extracted features from Deep CNNs with Support Vector Machine in the second, fusing deep features in the third, and employing PCA in the final experiment to reduce computational costs and feature vector size. This study [75] utilizes transfer learning, an AI technique, to improve DL models for medical imaging diagnosis. The input data is enhanced through pseudo-colored images with CLAHE and pixel-wise intensity correction. A novel hybrid processing method, LR-PCA, reduces multicollinearity in correlated deep data. High-performance accuracies of 98.60% and 98.80% are achieved on benchmark datasets, mini-MAIS and INbreast, respectively. The proposed [76] method employs four mammography imaging

datasets, each consisting of 1145 images of benign, normal, and malignant tissue, to build base classifiers. It utilizes an ensemble approach that adaptively blends decision scores of base models using the Gompertz function for fuzzy rankings. The suggested Inception V4 ensemble model achieved an impressive accuracy rate of 99.32%, making it a valuable tool for early breast cancer detection and timely medical intervention. The proposed [77] strategy combines supervised and consistency training for improved model resilience. It utilizes adaptive token sampling to select essential tokens from input images, resulting in enhanced performance. The technique is validated on histopathology and ultrasonography datasets, consistently outperforming CNN baselines in both learning tasks. The proposed [78] DecT model utilizes color deconvolution in its convolution layers to leverage histopathology image staining characteristics and employs a self-attention technique to match HED channel information. To address information loss during RGB to HED conversion, a residual connection-like technique combines RGB and HED images. By dividing the training procedure into two steps, the DecT model adapts deconvolution layer parameters effectively, achieving average accuracies of 93.02% on BreakHis, 79.06% on BACH, and 81.36% on UC datasets. In this study [79], data augmentation strategies for mammography image processing are discussed in detail. It examines several data augmentation methods used on mammography images, including simple and DL-based methods. The study presents cutting-edge techniques for image augmentation and offers insights regarding augmentation during testing. The paper makes the case that more research is necessary before drawing conclusions about how data augmentation affects the capacity of supervised DL paradigms to generalize information. The proposed [80] CAD system utilizes deep feature transfer learning, with Inception-ResNet-v2 demonstrating the best feature extraction capabilities among 16 pre-trained networks. By combining CatBoost, XGBoost, and LightGBM, the suggested strategy achieved high average accuracy (96.82%, 95.84%, 97.01%, and 96.15%) in all magnifications on the BreakHis dataset, which comprises 2,480 benign and 5,429 malignant histopathology images. The Table 2 summarizes various methods used in ML-based breast disease diagnosis along with their corresponding advantages:

4.1.2 Lung Cancer

The paper [81] explores lung cancer segmentation and classification using the Swin Transformer model. The pre-trained Swin-B model exhibits a 2.529% advantage over ViT in classification, achieving a top-1 accuracy of 82.26%, while the Swin-S model outperforms other methods in segmentation based on mean Intersection over Union (mIoU).

Table 2 Methods and Advantages in ML-Based Breast Cancer Diagnosis

Reference	Method	Advantage
[71]	AI-based CAD framework utilizing ViT's self-attention mechanism and CNN ensemble transfer learning for breast cancer diagnosis	Precise high-level deep features; Achieves high results for binary and multi-class prediction
[72]	Hybrid DL CAD system combining Residual Convolutional Network, Transformer Encoder, and MLP for breast cancer lesion classification	Effective discrimination between binary and multiclassification; High accuracy using pre-trained CNN and datasets
[73]	"Feature Score computation" method to re-rank Google search engine results based on page characteristics, enhancing corpus relevancy	Novel approach to enhance corpus relevancy; Increases relevance from 88–99%
[74]	Contemporary CAD architecture with DL and Support Vector Machine integration for breast cancer detection	Multiple experiment approaches for enhanced detection; Utilizes Deep CNNs and PCA for efficient computational cost
[75]	Transfer learning combined with AI techniques for medical imaging diagnosis	Pseudo-colored images and hybrid processing method enhance model performance; High accuracies achieved on benchmark datasets
[76]	Ensemble approach using Inception V4 model on mammography images for breast cancer detection	Adaptive blending of decision scores for accurate detection; Achieves impressive accuracy rate of 99.32%
[77]	Supervised and consistency training with adaptive token sampling for improved model resilience	Enhanced model resilience; Outperforms CNN baselines consistently on histopathology and ultrasonography datasets
[78]	DecT model utilizing color deconvolution, self-attention, and RGB-HED information fusion for histopathology image classification	Explores histopathology image characteristics effectively; Adapts deconvolution parameters with improved accuracy
[79]	Detailed discussion on data augmentation strategies for mammography images	Comprehensive examination of augmentation methods; Offers insights and perspectives on DL paradigm generalization
[80]	CAD system using deep feature transfer learning and ensemble methods for histopathology image classification	Utilizes multiple pre-trained networks for feature extraction; Achieves high accuracy on BreakHis dataset for both benign and malignant images

The study highlights the benefits of pre-training for improving the Swin Transformer model's accuracy across different tasks. The paper [82] introduces a DL system for computer-aided lung cancer diagnosis, employing 3D deep networks for automated detection and risk assessment of malignant nodules in 3D lung CAT scans. The study proposes [83], a 3D transformer model for detecting lung nodules within candidate regions using self-supervised learning. The model uses a 3D vision transformer to split CT image volumes into non-overlapping cubes, extract embedding features, and apply self-attention for prediction. Efficient training is achieved through unsupervised region-based contrastive learning on open CT images from the LIDC-IDRI dataset. They propose a pipeline [84], HViT4Lung, that combines Transformers and CNN models for improved computer-aided diagnosis of lung cancer using real lung CT scan data, achieving excellent classification accuracy and aiding in prompt treatment procedures by identifying lung nodules and assessing malignancy characteristics. The paper [85] presents an overview of AI applications in lung cancer detection and prognosis, covering reinforcement learning, ML, and DL techniques. It highlights AI's potential role in transforming lung cancer diagnosis and prognosis for precise patient treatment. This paper [86] is to discuss the different applications of AI approaches in lung cancer detection. The paper provides an overview of how AI can simplify tasks, reduce the effort required of radiologists, and increase the accuracy of nodule detection. The study [87] proposes a DL-based system for early lung cancer detection in CT scan imaging, comprising segmentation and classification components using self-supervised and UNETR networks, respectively. The method utilizes 3D-input CT scans and achieves remarkable segmentation (97.83%) and classification (98.77%) accuracies, indicating improved outcomes over prior state-of-the-art approaches. Table 3 explore the diverse landscape of ML-driven lung cancer diagnosis, showcasing cutting-edge methods and their significant advantages.

4.1.3 Brain Cancer

The study [88] focuses on brain tumor segmentation and classification methods, particularly highlighting ML, CNN, CapsNet, and ViT-based techniques. It comprehensively discusses the performance, contributions, challenges, and potential research directions of these advanced methods for brain tumor analysis. The study employs [89] DL architectures, including IVX16, to classify brain tumors through transfer learning. Various architectures are compared, with peak accuracies for VGG16, InceptionV3, VGG19, ResNet50, InceptionResNetV2, Xception, and IVX16 being 95.11%, 93.88%, 94.19%, 93.88%, 93.58%, 94.5%, and 96.94%, respectively, assessed using Explainable AI. The

Table 3 Methods and Advantages in ML-Based Lung Cancer Diagnosis

Reference	Method	Advantage
[81]	Lung cancer segmentation and classification using Swin Transformer model	Pre-trained Swin-B model achieves 2.529% advantage over ViT in classification; Swin-S model excels in segmentation based on mIoU
[82]	DL system with 3D CNN for lung cancer diagnosis in 3D lung CAT scans	Automated detection and risk assessment of malignant nodules; Improves diagnostic accuracy
[83]	3D transformer model for lung nodule detection using self-supervised learning	Utilizes 3D vision transformer for detecting lung nodules; Achieves efficient training through contrastive learning
[84]	HViT4Lung pipeline combining Transformers and CNN models for lung cancer diagnosis	Improved diagnosis accuracy; Identifies lung nodules and assesses malignancy characteristics
[85]	AI applications overview in lung cancer detection and prognosis	Highlights potential role of AI in transforming lung cancer diagnosis and prognosis for precise patient treatment
[86]	AI approaches in lung cancer detection	Simplifies tasks, reduces radiologist effort, increases nodule detection accuracy
[87]	DL-based system for early lung cancer detection in CT scans	Segmentation and classification components achieve remarkable accuracy; Outperforms prior state-of-the-art approaches

research introduces Vision Transformers, contrasts them with ensemble models and transfer learning, aiming to enhance brain tumor detection accuracy in MRI scans for professionals and laypeople. This research [90] introduces a QSM model for estimating brain magnetic susceptibility using UK Biobank imaging data, revealing links between genetic variations, brain conditions, and tea consumption. It enhances the neuroimaging database with QSM-derived estimates and provides insights into genetic influences on brain susceptibility. The study [91] introduces a novel approach employing DNN for data augmentation in 2D brain tumor segmentation within MR images. The proposed method, Znet, employs encoder-decoder structures, data amplification, and skip-connections to effectively segment brain tumors, achieving high performance metrics including dice similarity coefficient. This work underscores the potential of DL techniques, particularly the Znet framework, for accurate and automated brain tumor detection and segmentation in MR images. This paper [92] proposes an attention-based CNN for reliable brain tumor segmentation, achieving Dice similarity coefficients of 0.83, 0.86, and 0.90 for augmentation, core, and total tumors. The algorithm automatically learns the feature hierarchy, using the

BRATS'20 dataset with one target mask file and four MRI modalities. Leveraging long LSTM [93] networks to exploit spectrum feature temporal dependencies aids in reducing EEG noise and inter-subject variability during office-based anesthesia. The novel Anes-MetaNet framework combines CNN feature extraction, LSTM temporal modeling, and meta-learning to enhance performance, compared to existing techniques, on office-based anesthesia EEG data. Extensive experimentation [94] automating brain tumor detection via deep CNNs, transfer learning, and fine-tuning. The proposed model surpasses other CNNs in accuracy, weight sizes, achieving 98.87% classification accuracy, surpassing cutting-edge counterparts. In [95], a customized Gaussian Convolutional Neural Network (GCNN) for glioma grade and brain tumor type classification, encompassing grades 1 to 4 and tumor types pituitary, glioma, and meningioma. Employing diverse filters on initial brain tumor (BT) images pre-processing to eliminate noise and enhance image smoothness, subsequently amplifying classification efforts. A pioneering Hybrid-DANet [96] model is unveiled, which integrates a compactor-interpreter hybrid weights alignment alongside a multi-dilated attention network. This integration is specifically designed for the purpose of automating brain tumor segmentation. The approach incorporates a unique hybrid weight alignment with HWADA within skip connections, along with a MCS for improved channel-specific data extraction. Additionally, a residual module (RM) is applied to mitigate the vanishing gradient problem and enhance accuracy. Enhancing MRI image quality via a three-step preprocessing [97], a novel DCNN architecture integrates batch normalization for efficient training, featuring streamlined layers for computational efficiency. Performance evaluation on a dataset of 3394 MRI images demonstrates a noTable 98.22% accuracy, including 99% for glioma, 99.13% for meningioma, 97.3% for pituitary, and 97.14% for normal image identification. Table 4 explores the diverse landscape of ML-driven brain cancer diagnosis, showcasing cutting-edge methods and their significant advantages.

4.2 Kidney Disease

Fuzhe et al. [98] proposed a modified ANN (HMANN) for detecting, segmenting, and diagnosing chronic renal failure. It uses a combination of a SVM and MP with a Backpropagation algorithm and works based on preprocessed ultrasound images. The HMANN method has high accuracy in kidney segmentation and reduces the time for contour delineation. Alvaro et al. [99] investigated the use of ML techniques for early diagnosis of Kidney Disease in developing countries. The study compared different ML techniques. The study found that the J48 decision tree was the most suitable method for CKD screening in developing

Table 4 Methods and Advantages in ML-Based Lung Cancer Diagnosis

Reference Number	Method	Advantage
[88]	ML, CNN, CapsNet, ViT-based techniques	Comprehensive analysis of advanced brain tumor segmentation and classification methods
[89]	IVX16, transfer learning	High accuracy in brain tumor classification using DL architectures
[90]	QSM model	Genetic insights into brain magnetic susceptibility and its link to brain conditions and tea consumption
[91]	Znet framework, DNN	Accurate and automated brain tumor detection and segmentation in MR images
[92]	Attention-based CNN	Reliable brain tumor segmentation with high Dice similarity coefficients
[93]	Anes-MetaNet framework, CNN, LSTM	Improved office-based anes-thesia EEG data analysis using CNN and LSTM
[94]	Deep CNNs, transfer learning, fine-tuning	Automated brain tumor detection with high accuracy
[95]	Gaussian CNN (GCNN)	Customized model for glioma grade and brain tumor type classification
[96]	Hybrid-DANet model	Automation of brain tumor segmentation with improved alignment and attention network
[97]	DCNN, batch normalization, preprocessing	Enhanced MRI image quality and efficient training with high accuracy

countries with 95.00% accuracy, while the other methods showed moderate accuracy with more difficulty in interpretation. Segal et al. [100] aimed to create a predictive model for end-stage renal disease (ESRD) using ML algorithms using large-scale database of CKD. The authors employed a feature embedding method with the Word2Vec algorithm to capture temporal information for predictions. The analysis was performed using the gradient boosting tree algorithm. The author got a specificity of 95.8%. Sunil et al. [101] proposed an ML model to predict the final stage of kidney disease in patients with type 2 diabetes. The study used data from 11,789 patients from three clinical trials and used 18 baseline demographic. The best-performing ML model was the feed-forward neural network, which had an AUC of 0.82, 0.81, and 0.84. Yutong et al. [102] build a risk prediction model for End-Stage Renal Disease (ESRD) resulting from Diabetic Kidney Disease (DKD) in patients with Type 2 Diabetes Mellitus (T2DM). Four ML algorithms were used to identify important clinical and pathological features. The RF algorithm performed the best in predicting ESRD with the highest accuracy of 82.65%. Elias et al.

[103] aimed to build a tool for predicting Chronic Kidney Disease (CKD) occurrence using ML techniques. They balanced the classes to address the non-uniform distribution of instances, performed feature ranking and analysis, and trained and evaluated various ML models. The results showed that the Rotation Forest (RotF) was the best performer with an AUC of 100. The authors in [104] focused on using a CKD dataset which is publicly available. They used three different algorithms to train three models for CKD prediction. The results showed that the LR algorithm achieved a higher accuracy rate of 97% compared to DT which had 96.25% accuracy and k-NN which had 71.25% accuracy. This study [105] examined a publicly available dataset of healthy and kidney disease patients to develop and analyze different predictive models for kidney disease detection. Several different ML algorithms used in combination with two feature selection techniques, namely RFE and Chi-Square tests. The results showed that the logistic regression based prediction model achieved the highest accuracy of 98.75% using the best features selected by Chi-square technique. H. Polat et al. [106] proposed the use of a SVM to predict renal disease. They achieved an accuracy rate of 97.5%, which they then improved by 1% by employing various feature selection methodologies. Almansour et al. [107] developed a technique to prevent chronic kidney disease using ML classification methods such as SVM and ANN. While SVM had an accuracy rate of 97.6%, ANN achieved a higher accuracy rate of 99.75%. This suggests that ANN is a better method for predicting chronic kidney disease compared to SVM. The paper presents a novel method for assembly clone search [108], leveraging large-scale pretrained natural language models through transfer learning (TL). By incorporating a reinforcement learning agent and a Variational Information Bottleneck learning strategy, it overcomes limitations, achieving enhanced generalization for unseen architectures. Simulation results demonstrate the system's effectiveness in clone search scenarios compared to existing solutions. In [109], the author introduces an automated DL-based approach for predicting kidney organoid differentiation status from bright-field images using a CNN. The study shows the CNN's accurate recognition of differentiation status and outperforms human-based classifiers. The study introduces a novel method [110] for cultivating human pluripotent stem cell-derived kidney organoids using a dECM hydrogel. Kidney organoids with dECM display more mature glomerular growth patterns and successfully model Fabry nephropathy with vasculopathy. In [111], a novel DL model is introduced, which combines surface and section views of kidney stone fragments from endoscopic images, resulting in improved discrimination performance. The study employs a two-step transfer learning approach to train a multi-branch model, enhancing the

model's generalization capabilities. The paper presents a DL technique [112] for predicting kidney CT-scan images, detecting kidney abnormalities (stones, cysts, and tumors). It employs CNN with data augmentation, normalization, and transfer learning to enhance accuracy. The balanced dataset achieved an impressive accuracy and F1 score of around 99%, indicating potential value in aiding medical professionals, especially in remote areas with limited access to specialists. The paper presents [113] an AI-based kidney diseases auto-diagnostic system using 12,446 CT images for kidney stones, cysts, and tumors. Six ML models, including variants of VT (EANet, CCT, Swin) and DL models (Resnet, VGG16, Inception v3) with modified last layers, were evaluated. The proposed model achieved the highest, and VGG16 exhibited promise in monitoring essential anatomy abnormalities. A DL method [114] is developed to estimate glomerular filtration rate (eGFR) and chronic kidney disease (CKD) status from kidney ultrasound images. The neural network employs a transfer learning approach

with a ResNet model pretrained on an ImageNet dataset and utilizes kidney length annotations for cropping and data augmentation. Bootstrap aggregation is applied to reduce overfitting and improve model generalizability, enabling the deep neural network to determine CKD status based on eGFR values below 60 ml/min/1.73 m². The study [115], compares individualized models with global and subgroup models to assess AKI risk variability in different subpopulations. Leveraging transfer learning techniques and meta-regression, the personalized model demonstrates superior predictive performance over the global model, as indicated by the AUROC, for general inpatients and high-risk/low-risk subgroups. The paper presents a predictive CKD model [116] using diagnostic medical data from the UCI repository. Performance tuning involves hyperparameter adjustment and ensemble weighting (Ranking Weighted Ensemble), resulting in impressive metrics: 98.75% accuracy, 100% sensitivity, 96.55% specificity, and 99.03% F1 score, indicating its potential for automated CKD severity detection. The Table 5 highlights a collection of advanced methods and techniques used for predicting and diagnosing various aspects of kidney diseases, including early detection, organoid differentiation, and risk assessment.

Table 5 Advances in Kidney Disease Prediction and Diagnosis

Reference Number	Method Name	Advantage
[98]	HMANN for chronic renal failure	High accuracy in kidney segmentation
[99]	ML techniques for early diagnosis	J48 decision tree suitable for CKD screening
[100]	Predictive model for ESRD	Specificity of 95.8%
[101]	ML model for final stage prediction	Feed-forward neural network with high AUC
[102]	Risk prediction for ESRD from DKD	RF algorithm predicts ESRD with 82.65% accuracy
[103]	Tool for predicting CKD occurrence	Rotation Forest with AUC of 100
[104]	CKD prediction using different ML algorithms	LR achieved highest accuracy of 97%
[105]	Predictive models for kidney disease	Logistic regression with highest accuracy
[106]	SVM for renal disease prediction	Improved accuracy using feature selection
[107]	SVM vs. ANN for chronic kidney disease	ANN achieves higher accuracy
[108]	Transfer learning for clone search	Enhanced generalization for unseen architectures
[109]	DL-based kidney organoid prediction	CNN outperforms human-based classifiers
[110]	Kidney organoid cultivation method	Improved glomerular growth patterns
[111]	DL model for kidney stone detection	Improved discrimination performance
[112]	DL for kidney CT-scan images	Impressive accuracy and F1 score
[113]	AI-based auto-diagnostic system	High accuracy in detecting kidney abnormalities
[114]	DL for eGFR and CKD status estimation	Neural network estimates CKD status

4.3 Heart Disease

Jian et al. [117] propose an efficient and accurate heart disease diagnosis system based on ML techniques. They have also used different feature selection algorithms, as well as a novel Fast Conditional Mutual Information Feature Selection Algorithm, to improve the accuracy and reduce the execution time of the system. The simulations results show that the proposed FCMIM-SVM system achieved good accuracy compared to previous methods. Fajr et al. [118] proposed a framework to classify ECG signals. The author evaluated several ML classification algorithms in Spark-Scala on two datasets. The results showed that the GDB Tree algorithm had an overall accuracy of 96.75% for binary classification, while the random forest algorithm achieved an accuracy of 97.98% for 2-class classification and 98.03% for 3-class classification. In [119] authors applied a combination of neural networks and dimensionality reduction techniques to classify electrocardiogram (ECG) records. They tested their approach on the arrhythmia database, using a test set consisting of 18 ECG records, each 30 minutes long. The classification performance achieved an accuracy of 96.97%. In this research paper by [120], an investigation into heart disease is conducted through the utilization of supervised learning algorithms. The dataset employed is sourced from the Cleveland database of the UCI repository and encompasses 303 instances and 76 attributes; however, for testing purposes, only 14 attributes were

taken into account. The primary objective revolves around forecasting the potentiality of heart disease development. Notably, the KNN algorithm yielded the highest level of accuracy in this context. Abid et al. [121] proposed a framework for cardiovascular patient survivors using nine ML algorithms. The author used a dataset of 299 patients of heart failure survival. Experimental results prove that the Extra tree classifier gives us an accuracy of 92.62% using SMOTE technique. Melillo et al. [122] proposed a classifier for patients with high and low risk of cardiovascular disease. The classifier was based on the CART method and showed good results with 93.3% sensitivity and 63.5% specificity. However, the study was limited in size and the results need to be tested on a larger dataset for effectiveness. In [123], an AI-based heart detection system was developed using ML techniques. The system underwent data processing, including the handling of categorical variables. A random forest classifier algorithm was used for data analysis, resulting in an accuracy rate of approximately 83% with room for further improvement. Ghulab et al. [124] examines the performance of various ML algorithms on two datasets of cardiac data. The GBC algorithm was used in combination with GridSearchCV to optimize performance. The findings from this study revealed that the GBC algorithm, coupled with GridSearchCV for hyperparameter tuning, achieved the most noteworthy and almost equivalent test and training accuracies, attaining 100% and 99.03%, respectively, across both datasets. Additionally, it was observed that the XGBoost classifier, operating without GridSearchCV, demonstrated the highest and nearly similar test and training accuracies, reaching 98.05% and 100%, respectively. Olaniyan et al. [125] proposed a three-step approach to accurately predicting angina via a neural network technology, with a success rate of 88.89%. Andres et al [126] proposed a heart disease prediction model using Hungarian Cleveland dataset. The author used several ML algorithms for prediction and used PCA for feature selection and dimensionality reduction. The main objective [127] is to build a heart disease prediction model using AI and ML techniques. The paper introduces a hybrid Faster R-CNN with SE-ResNet-101 model and HLDA-MALO model for sensor data and image classification, respectively, establishing a strong and effective framework for heart disease prediction. In this research [128], we perform correlation and inter-dependence analysis on diverse medical features related to heart disease. A filter-based feature selection method is utilized to identify the most relevant features for heart disease detection. We explore multiple ML classification models with complete and reduced feature subsets as inputs, evaluating their performance using Accuracy, ROC curve, and F1-Score. The paper proposes a lightweight neural network [129] for real-time ECG anomaly detection in

wearable IoT Edge sensors, combining LSTM cells and MLP for complementary feature learning. The network achieves 97% accuracy on the MIT-BIH arrhythmia database, expanded using SMOTE for training imbalance. After mapping to fixed-point model, retraining, and porting to ARM Cortex M4, it exhibits 50% power savings in laboratory tests by gating wireless transmission, making it suitable for IoT wearables with limited connectivity needs. The paper [130] presents a smart healthcare monitoring system utilizing DL and IoT devices for accurate heart disease risk prediction. Employing Bi-LSTM, a variant of DL, the system achieves high accuracy, and outperforming existing heart disease prediction systems. The study highlights DL's potential, particularly RNNs like Bi-LSTM, in effectively handling long-term dependencies in temporal event analysis for medical predictive analytics. The paper [131] introduces a novel framework utilizing deep neural networks (DNNs) for heart disease prediction. It incorporates enhanced cluster-based BiLSTM classification to achieve accurate heart disease incidence predictions with a remarkable 91% accuracy rate, effectively identifying heart risk factors. The paper introduces 'Ensem-HAR' [132], an ensemble of four DL-based technique designed for smartphone sensor-based human activity recognition in elderly health monitoring. The proposed model achieves impressive accuracy across three benchmark datasets, demonstrating its efficacy in accurately recognizing human activities. In [133], the study addresses challenges of imbalanced data in existing methods, impacting practicality and effectiveness. It highlights drawbacks like biased performance towards the majority class and limited model interpretability. The paper emphasizes analyzing ML performance on imbalanced heart disease data, particularly in less developed countries with financial constraints and limited healthcare access, necessitating affordable and reliable diagnoses. The paper presents a novel hybrid algorithm [134], combining MLP and PSO, for heart disease detection, outperforming ten other ML algorithms with an accuracy of 84.61%. The proposed MLP-PSO algorithm has the potential to enhance early and accurate cardiac disease diagnosis, improving patient care and optimizing valuable face-to-face time with patients. In [135], the study explores multiple data-driven models to predict future heart rates (HR) from accelerometer data of healthy individuals. The ARIMA model employing walk-forward validation and linear regression exhibited superior performance across all durations. On the other hand, alternative models demonstrated excellence particularly for durations extending beyond 1 min. This research underscores the importance of data analytics in accurately predicting HR using accelerometers, benefiting IoT and wearable healthcare monitoring systems for timely HR prediction. The healthcare framework employs Blockchain

technology [136] with on-chain and off-chain schemes. It undergoes extensive performance evaluation through multi-hosted testbed, realistic GB traffic analysis using Tcpdump tools, and employs Hyperledger Fabric 2.0 smart contracts with unified performance analysis through Hyperledger Caliper 0.4.2. The prototype outperforms existing methods, surpassing prevalent use of SOLO, exhibiting high throughput, low latency, and efficient resource utilization for healthcare Blockchain transaction handling. The Table 6 presents cutting-edge methods in heart disease prediction and diagnosis, showcasing advancements in accuracy and efficiency achieved through ML, DL, and novel techniques.

4.4 Diabetes

Roshan et al. [137] tried to focus on the diagnosis of diabetes using ML models. The accuracy achieved was 86% for Gradient Boosting, 79% for Logistic Regression, and 77% for Naive Bayes. The authors of the study [138] used both a multilayer network trained by the Levenberg–Marquardt algorithm and a probabilistic neural network to diagnose diabetes. They compared their results to previous studies using the same dataset from the UCI repository and found that the neural network (ANN) was effective in diagnosing Pima-diabetes. In [139], the author used a logistic regression model in predicting the risk value of diabetes based on the P value and odd ratio. The algorithms were evaluated using different metrics under three data partitioning protocols (K2, K5, and K10). The author implemented four ML algorithms for the prediction of a diabetes patient. The study [140] compared five ML-based prediction models for the prediction of undiagnosed Type 2 Diabetes Mellitus (T2DM). The performance was measured by the prediction of fasting plasma glucose level using 100 bootstrap iterations with increasing amounts of data over 6-month intervals. The results showed that the simple regression model had the lowest average RMSE of 0.838. The authors in this study [141] have proposed an ensemble ML approach for the detection of diabetes patients. This ensemble uses three algorithms with soft voting classifier. The performance of the proposed methodology is evaluated using different metrics. The method is tested on PIMA diabetes datasets and compared to conventional ML algorithms. In this study [142], data on type 2 diabetes was gathered from a hospital in Kano, Nigeria and used to create predictive models using various ML algorithms. The random forest algorithm showed the highest accuracy with 88.76%, and the random forest and gradient boosting models were found to have the best model in terms of ROC, with 86.28% respectively. Aishwarya et al. [143] proposed a new diabetes prediction model that considers additional factors besides the typical ones like glucose levels, body mass index (BMI), age, and

insulin. The model is based on a larger, more comprehensive dataset and is intended to improve the accuracy of diabetes classification compared to existing models. The study also introduces a pipeline model for diabetes prediction that aims to further enhance the accuracy of the classification. The research [144] studied the use of three supervised ML models for classifying the Pima Indians diabetes dataset. The results showed that a Naive Bayes model works well with a refined feature selection, while the Random Forest model works better with more features. Chollette et al. [145] studied the effects of a deep neural network model for accurately predicting diabetes mellitus and determining its severity during diagnosis. Using the PIMA Indian dataset, the model achieved a train and test prediction accuracy of 99.01% and 97.25%, respectively. When tested on the LMCH dataset, these figures increased to 99.57% and 97.33%. In [146], PCA was used for feature selection, and the median was used for missing value imputation. MLP classification approach was then used, resulting in an accuracy rate of 75.7%. Singh et al. [147] utilized a combination of Linear SVM, Radial Basis function SVM, Decision Tree, and K-Nearest Neighbor algorithms to classify data, resulting in an accuracy rate of 83.8%. In [148], the author presents a Source-Free TL (SFTL) model for referable diabetic retinopathy (DR) screening, addressing limited annotated data in medical image analysis. The SFTL model utilizes a generator and discriminator with adversarial and target reconstruction losses, alongside feature consistency loss, yielding competitive results on the APTOS 2019 dataset, outperforming other supervised models. In this study [149], A novel two-step framework is introduced for diabetic retinopathy (DR) diagnosis and disease stage screening, employing ML models. The first step involves identifying essential characteristics for precise diagnosis, and the second step selects a reliable ML classification method. Ensemble learning is employed to handle heterogeneous and unbalanced databases, yielding a robust ML model capable of recognizing different DR stages with high accuracy, F1-score, precision, specificity, sensitivity, and AUC. Comparisons with other ML models are also presented by the author. The study presents an automated diagnostic method [150] for early-stage Diabetic Retinopathy (DR) diagnosis using retinal fundus images. The approach utilizes the KMA-optimized ABiLSTM technique, successfully detecting and classifying DR irregularities. Comparative evaluations demonstrate the superiority of the proposed method with an impressive result and reduced computation time (50 s), offering a promising solution for DR staging challenges. In this study [151], an automated DR detection and classification method is proposed, utilizing U-Net and DL techniques to overcome limitations of traditional hand-crafted feature extraction methods. The methodology

Table 6 Innovations in Heart Disease Prediction and Diagnosis

Reference Number	Method Name	Advantage
[117]	Efficient heart disease diagnosis (ML)	Novel Fast Conditional Mutual Information Feature Selection Algorithm to improve accuracy and reduce execution time.
[118]	ECG signal classification (ML)	GDB Tree algorithm achieves 96.75% accuracy for binary classification; random forest reaches 97.98% (2-class) and 98.03% (3-class).
[119]	ECG classification with neural networks (ML)	Achieves 96.97% accuracy in classifying electrocardiogram records using neural networks and dimensionality reduction techniques.
[120]	Heart disease prediction (Supervised ML)	KNN algorithm yields high accuracy in forecasting heart disease development using selected attributes.
[121]	Cardiovascular patient survival (ML)	Extra tree classifier achieves 92.62% accuracy for cardiovascular patient survival using SMOTE technique.
[122]	Risk classification for cardiovascular disease (ML)	CART-based classifier shows 93.3% sensitivity and 63.5% specificity for high/low-risk cardiovascular disease patients.
[123]	AI-based heart detection system (ML)	Random forest classifier achieves an accuracy of approximately 83% for detecting heart disease using AI and ML techniques.
[124]	Performance analysis of ML algorithms (ML)	GBC algorithm coupled with GridSearchCV achieves noteworthy and almost equivalent test and training accuracies.
[125]	Angina prediction via neural network (ML)	Achieves 88.89% success rate in accurately predicting angina using a neural network technology.
[126]	Heart disease prediction using Hungarian dataset (ML)	Uses PCA and ML algorithms for heart disease prediction, showcasing effectiveness.
[127]	Hybrid Faster R-CNN and HLDA-MALO model (DL/ML)	Hybrid model for sensor data and image classification establishes a strong framework for heart disease prediction.
[128]	Feature selection and ML classification (ML)	Utilizes correlation analysis and filter-based feature selection for heart disease detection, evaluates ML classification models.
[129]	Real-time ECG anomaly detection (DL)	Lightweight neural network combining LSTM and MLP achieves 97% accuracy in real-time ECG anomaly detection.
[130]	Smart healthcare monitoring system (DL/IoT)	Utilizes Bi-LSTM for accurate heart disease risk prediction in IoT devices, highlighting potential of RNNs in medical analytics.
[131]	Cluster-based DNN for heart disease prediction (DL)	Enhanced BiLSTM classification achieves 91% accuracy in predicting heart disease incidence and identifying risk factors.
[132]	Ensemble-based smartphone activity recognition (DL)	'Ensem-HAR' ensemble achieves impressive accuracy in recognizing human activities using smartphone sensor data.
[133]	ML performance on imbalanced heart disease data (ML)	Addresses challenges of imbalanced data, emphasizes reliable ML performance in less developed countries with limited healthcare access.
[134]	Hybrid MLP-PSO algorithm for heart disease detection (ML)	Novel hybrid algorithm combines MLP and PSO, achieving 84.61% accuracy in cardiac disease diagnosis.
[135]	Predicting future heart rates from accelerometer data (Data-driven)	ARIMA model and alternative approaches predict heart rates with varying accuracies, benefiting IoT healthcare monitoring systems.
[136]	Healthcare framework with Blockchain technology (Blockchain)	Utilizes Blockchain for healthcare transaction handling, showcasing high throughput, low latency, and efficient resource utilization.

follows a two-stage approach. Image processing techniques are applied to address the low percentage of positive cases in OD and BV detection, enhancing image quantity and quality. The proposed methodology efficiently detects diabetic retinopathy (DR) using retinal biomarkers (MA, HM, and EX) with high accuracies: 97.92% on EyePACS-1, 94.59% on Messidor-2, and 93.52% on DIARETDB0 datasets. In this study [152], the authors aim to address diabetic retinopathy (DR) by proposing a novel modified GC deep belief network utilizing the dwarf mongoose optimization algorithm for effective DR prediction and classification. The

approach achieves an impressive 97% accuracy in classifying fundus retinal images into normal and abnormal categories, demonstrating its efficacy on the ODIR-2019 dataset. Timely DR identification is vital to prevent complete blindness caused by retinal damage. This paper introduces a novel contribution [153] to radiomics, applying computer vision techniques to (OCT, OCTA) images for DR diagnosis. ML classifiers demonstrate effective identification of patients. The addition of clinical variables enhances accuracy, showcasing radiomics-based ML techniques' potential for diabetic retinopathy screening in type 1 DM patients.

The paper conducts an extensive investigation of DR grading protocols and taxonomy [154], identifies relevant DL-related keywords, and formulates a search string with “AND” or “OR” Boolean operators. DL algorithms and methods for DR classification are analyzed and compared, considering publicly available datasets for empirical understanding. Commonly used algorithms for DR classification are identified in this study. This study [155] assesses ML prediction models’ effectiveness in detecting VTDR in type 2 diabetes mellitus (DM) patients using medical data from a data warehouse. ML approaches are utilized to create VTDR prediction models achieving strong F1 scores up to 0.81, 0.70, and 0.66 on the test set, enabling accurate VTDR forecasting over ten years. The proposed models facilitate timely referral of high-risk subjects to ophthalmologists from primary care, aiding in early diagnosis and effective management of diabetic retinopathy, reducing the risk of vision loss in type 2 DM patients. In [156], the authors propose a novel approach for diabetic retinopathy classification using OCTA vascular density maps and SVM classifier optimized by a genetic evolutionary algorithm. The SVM model achieves 90% accuracy in distinguishing between PDR and NPDR based on OCTA vascular density maps of DCP, indicating the potential of this method for accurate diagnosis and classification of diabetic retinopathy patients. This study opens new avenues for future research in this field. The paper presents a novel DCNN [157] tailored for DR prediction in retinal fundus images, incorporating correlations between long-range patches (LRP) in the DL framework. This approach significantly enhances DR detection by improving local patch features, crucial for identifying DR lesions, and the residual structure allows easy integration into other networks. The suggested method outperforms existing models on the Messidor and EyePACS datasets, emphasizing the importance of considering both local and long-range global dependencies for effective DR detection in retinal fundus images. Table 7 Explore the evolution of methods for diabetic retinopathy diagnosis and prediction, showcasing their unique advantages and contributions to improved healthcare outcomes.

4.5 COVID-19

Sohaib et al. [158] used DCNN to diagnose COVID-19 pneumonia with the goal of maximizing accuracy. The study proposed the use of the Inception V3 model with transfer learning and achieved a validation accuracy of 93%. In another study, Sohaib et al. [159] propose a DL-based system for COVID-19 detection. They evaluate the performance of several pre-trained DL models and propose a CNN architecture for the reliable classification of X-ray images. The data used in the study consists of 2,541 X-ray images

from COVID-positive and healthy patients. They achieved highest accuracy rate of 99.68%. This study [160] aimed to classify lung CT images for early diagnosis of Covid-19 using DL techniques. A CNN was implemented on a dataset comprising 1,396 lung CT images, consisting of 386 instances related to Covid-19 and 1,010 instances representing non-Covid-19 cases. The attained outcomes were subsequently compared with the results produced by conventional methods. Sara et al. [161] conducted a comparative analysis of prominent DL frameworks with the aim of automating COVID-19 classification via chest X-ray and CT images. A selection of Deep CNN was made, and the derived features were supplied to diverse ML classifiers for the classification task. Notably, the most favorable performance was observed when utilizing the DenseNet121 feature extractor in conjunction with the Bagging tree classifier, yielding a remarkable accuracy of 99%. Barstugan et al. [162] proposed a ML approach for COVID-19 classification based on CT images. They used 150 CT images and extracted patches with different sizes. They used various hand-crafted features such as GLCM, LDP, GLRLM, GLSZM, and DWT, and fed the extracted features into SVM classifier. The study used different cross-validations and found that the best accuracy of 98.77% was achieved by using the GLSZM feature extractor with 10-fold cross-validation. This paper [163] presents a method for early diagnosis of COVID-19 using X-ray images with the help of DL pre-trained models. The best model was determined by considering different model hyper parameters. The evaluation of the models was conducted using publicly accessible chest X-ray images, revealing that the most favorable outcomes were achieved by the ResNet-34 model, achieving an impressive accuracy of 98.33%. Wang and Wong [164] introduced a CNN model named COVID-Net, designed for the detection of COVID-19. The achieved testing accuracy of the model was notably high at 92.6%. In this study [165], chest X-ray scans captured in the Posterior-Anterior (PA) view were subjected to analysis using DL-based models. This investigation encompassed both COVID-19 positive and healthy patients. The dataset underwent preprocessing and augmentation, followed by a comparative assessment of three distinct models in terms of accuracy. Remarkably, the Xception model demonstrated the most superior performance, achieving an impressive accuracy of 97.97% in detecting chest X-ray images. In [166], the author introduced the LW-CORONet model, meticulously engineered to extract meaningful features from chest X-ray (CXR) images. A rigorous evaluation was conducted utilizing two extensive CXR datasets, yielding classification accuracies of 98.67% and 99.00% for multi-class scenarios, and 95.67% and 96.25% for binary classification cases, respectively. Comparative analysis was undertaken against four pre-trained CNN models and

Table 7 Advancements in Diabetic Retinopathy Diagnosis and Prediction

Reference Number	Method Name	Advantage
[137]	ML models for diabetes diagnosis	Achieved accuracies of 86% (Gradient Boosting), 79% (Logistic Regression), and 77% (Naive Bayes) in diabetes diagnosis.
[138]	Neural Network (ANN) for Pima-diabetes diagnosis	Effective diagnosis of Pima-diabetes using ANN and comparison with previous studies, highlighting ANN's diagnostic capabilities.
[139]	Logistic Regression for diabetes risk prediction	Utilized logistic regression model for diabetes risk prediction, evaluated under different data partitioning protocols.
[140]	Prediction models for undiagnosed T2DM	Compared five ML prediction models for undiagnosed Type 2 Diabetes Mellitus prediction, highlighting performance using RMSE.
[141]	Ensemble ML approach for diabetes detection	Proposed ensemble approach using three algorithms with soft voting classifier for diabetes detection, evaluated and compared.
[142]	ML algorithms for diabetes prediction in Nigeria	Random Forest algorithm achieves highest accuracy of 88.76% for type 2 diabetes prediction using data from Nigeria.
[143]	Improved diabetes prediction model	New model considers additional factors for diabetes classification, introduces pipeline model for further accuracy enhancement.
[144]	ML models for Pima Indians diabetes classification	Naive Bayes model with refined feature selection, Random Forest with more features; discussed models' performance.
[145]	Deep neural network for diabetes prediction	Deep neural network achieves high prediction accuracy of 97.25-99.57% for diabetes mellitus severity using PIMA Indian dataset.
[146]	Diabetes diagnosis with PCA and MLP	PCA for feature selection, median for missing value imputation, MLP classification approach with 75.7% accuracy.
[147]	Classification using SVM and Decision Tree	Combination of Linear SVM, RBF SVM, Decision Tree, and K-Nearest Neighbor for diabetes classification with 83.8% accuracy.
[148]	Source-Free TL model for diabetic retinopathy	Proposed Source-Free TL (SFTL) model achieves competitive results for referable diabetic retinopathy screening using APTOS 2019.
[149]	Two-step framework for DR diagnosis and screening	Novel framework with essential characteristics identification and ML classification for DR stages recognition with high accuracy.
[150]	Automated DR diagnosis using retinal images	KMA-optimized ABiLSTM technique for early-stage Diabetic Retinopathy (DR) diagnosis achieves impressive results and reduced time.
[151]	Automated DR detection using U-Net and DL	Two-stage methodology with U-Net and DL techniques for diabetic retinopathy detection, achieving high accuracies on datasets.
[152]	Modified GC deep belief network for DR prediction	Innovative deep belief network achieves 97% accuracy in classifying fundus retinal images for diabetic retinopathy prediction.
[153]	Radiomics-based ML techniques for DR screening	Computer vision techniques applied to OCT and OCTA images with clinical variables enhance DR screening accuracy.
[154]	Analysis and comparison of DL algorithms for DR	Comprehensive investigation and comparison of DL algorithms for diabetic retinopathy classification using various datasets.
[155]	ML prediction models for VTDR in type 2 DM	Effective ML models for detecting VTDR in type 2 diabetes mellitus patients using medical data, enabling accurate VTDR prediction.
[156]	SVM classifier with genetic algorithm for DR	Novel approach using OCTA vascular density maps achieves 90% accuracy in distinguishing PDR and NPDR in diabetic retinopathy.
[157]	DCNN with long-range patches for DR prediction	DCNN with long-range patch correlations significantly improves DR detection in retinal fundus images over existing models.

state-of-the-art counterparts. Importantly, the LW-CORONet model demonstrates a pronounced advantage by demanding fewer parameters and consuming less memory space compared to alternative models. The D3SENet [167] is a novel hybrid deep feature extraction network proposed to extract the most important features from chest x-ray images. A large and balanced dataset was used to classify Covid-19 infected, viral pneumonia infected and healthy individuals. The proposed method had an impressive result, achieving an accuracy of 98.48% on a three-class classification task. Weiha et al. [168] performed experiments on a dataset of 5143 X-ray images of three different classes (Healthy, Covid-19, and Pneumonia) using both SVM and CNN. Using handcrafted features such as Texture, GLCM, GLDM, FFT, Wavelet Transform and LBP, an SVM classifier was able to achieve an accuracy of 96.3%. When using an end-to-end CNN model, an accuracy of 98.3% was reached. The paper [169] provides a comprehensive analysis of COVID-19 mRNA vaccines, encompassing structural features, antigen design methods, and scalability. It discusses challenges and future prospects for preventive mRNA vaccines against infectious diseases, aiming to enhance understanding and their potential in epidemic control and public health protection. In [170], emphasizes the need for long-term follow-up studies to examine respiratory symptoms and lung damage in COVID-19 survivors. It underscores the significance of post-acute COVID-19 syndrome as a potential concern for healthcare providers, necessitating comprehensive management of chronic complications in the growing population of SARS-CoV-2-infected individuals globally. In [171], the investigation focuses on assessing the effectiveness of a COVID-19 booster dose vaccination in countering the Omicron variant. The study aims to clarify the ongoing debate on the importance of booster doses and their protective duration against the Omicron variant. Utilizing PubMed (Medline) for systematic search, the paper presents findings from various studies, indicating that a booster dose can increase neutralization titers against the Omicron variant and lower the risk of symptomatic breakthrough infections. In the study [172], non-invasive medical imaging is proposed to overcome RT-PCR limitations in COVID-19 diagnosis. Deep TL-based CNN approaches are evaluated to effectively analyze chest CT images, achieving improved COVID-19 screening with custom pre-trained models using batch normalization and regularization to address overfitting and enhance detection accuracy. The proposed DQ-NNPP method [173] combines two techniques with Privacy Preservation, offering improved data transmission security with reduced encryption and decryption time. By integrating cryptographic and biometric systems with DL techniques, it achieves authentication, anomaly detection, and enhanced medical system security.

Comparative analysis with standard approaches, such as SAB-UAS, MSCryptoNet, and PPDP, demonstrates its effectiveness. This paper introduces a novel ensemble method [174] for the detection of COVID-19 infection from lung CT scan images. The method involves fine-tuning multiple deep neural networks. By integrating the trained models, a robust ensemble classifier is constructed, outperforming existing methods and achieving superior outcomes. In [175], the proposed method is an ensemble approach, ET-NET, for rapid COVID-19 screening using three TL models to enhance supervised image classification performance. ET-NET combines base classifiers based on confidence scores, reducing prediction errors' variability and achieving superior results compared to the current technique on the same dataset by 1.56% through 5-fold cross-validation. The research [176] evaluates TL-based ResNet models for COVID-19 Pneumonia detection in Chest X-ray images, emphasizing the importance of CT/CXR in identifying COVID-19 infection. The study shows performance improvements in diverse target domains, attributing the success of TL. ResNet32, with a top-2 smooth loss function and cost-sensitive features, is employed as their DTL model to handle noisy and unbalanced data. Introduces [177] a novel weights-only transfer learning (TL) technique for automatic COVID-19 infection detection in CT-scan images on point-of-care devices. This approach prunes less important filters from convolutional layers, reducing pre-trained model weight parameters. Evaluation with the SARS-CoV-2 dataset confirms its effectiveness in reducing runtime resource requirements, making it suitable for point-of-care device implementation. In [178], The author suggests a transfer learning (TL) technique involving the fine-tuning of the ResNet50 model to classify COVID-19 patients. Ten pre-trained weights from diverse datasets are used, employing supervised and self-supervised learning. The COVID-19 Radiography Database is used for training and testing, with data transformation applied and the dataset split into train and test sets. The best-performing model among various fine-tuned ResNet50 TL models is identified, achieving binary classification for Covid and Normal categories. Table 8 Explore the latest developments in COVID-19 diagnosis and detection using DL techniques, highlighting the advantages and contributions of each method towards accurate and efficient screening.

5 Publicly Available Datasets

Data is the backbone of ML and DL models, and its availability is critical for training effective and accurate algorithms. Here's how data is useful in ML and DL:

- **Model Training:** Data is used to train ML and DL models to learn patterns and relationships between input features and target outputs. The more diverse and representative the data, the better the model can generalize to new, unseen examples.
- **Feature Learning:** In DL, large amounts of data help in learning hierarchical representations of features. These learned features capture meaningful patterns and enable the model to understand complex relationships between different data elements.
- **Generalization:** Adequate data helps models generalize well to unseen examples. When trained on diverse data,

ML and DL models can better handle variations and uncertainties present in real-world data, making them more robust and accurate.

- **Transfer Learning:** Pre-trained models, trained on large datasets, act as a starting point for transfer learning. They provide learned representations that can be fine-tuned on specific tasks, allowing for quicker convergence and improved performance with limited data.
- **Avoiding Overfitting:** Sufficient data helps in preventing overfitting, where the model becomes too specific to the training data and fails to generalize to new data. Adequate data helps regularize the model and make it more generalizable.
- **Data Augmentation:** When dealing with restricted data availability, data augmentation techniques prove valuable by artificially expanding the dataset through transformations such as rotations, flips, and color adjustments. This augmented dataset contributes to bolstering the model's resilience.
- **Data Bias and Fairness:** Datasets with sufficient diversity can help in mitigating bias in ML and DL models. By including representative samples from various demographics, the model can be trained to make fair and unbiased predictions.
- **Handling Noisy Data:** ML and DL models can learn to handle noisy data by exposing them to a wide range of examples. This exposure helps the model identify patterns even in the presence of noise.
- **Unsupervised Learning:** In unsupervised learning, data is crucial for clustering and discovering underlying structures and patterns within the data without labeled outputs.
- **Real-World Applications:** Ultimately, data is essential because ML and DL are practical tools that aim to solve real-world problems. The availability of relevant and diverse data ensures that the models are capable of delivering valuable insights and predictions in various domains, ranging from healthcare and finance to image and speech recognition.

Table 8 Advances in DL for COVID-19 Diagnosis and Detection

Ref. No.	Method Name	Advantage
[158]	DCNN for COVID-19 Pneumonia	Achieved 93% validation accuracy using Inception V3
[159]	DL-Based COVID-19 Detection	Attained 99.68% accuracy with proposed CNN architecture
[160]	CNN for Early COVID-19 Diagnosis	Successfully classified lung CT images for diagnosis
[161]	DL Framework Comparison	Best performance using DenseNet121 with Bagging tree
[162]	ML Approach for CT Classification	Achieved 98.77% accuracy using GLSZM feature extraction
[163]	DL-Based Early COVID-19 Diagnosis	ResNet-34 achieved 98.33% accuracy on X-ray images
[164]	COVID-Net CNN Model	Achieved testing accuracy of 92.6% for COVID-19 detection
[165]	Xception CNN for Detection	Highest accuracy of 97.97% in detecting chest x-rays
[166]	LW-CORONet Feature Extraction	High classification accuracies with fewer parameters
[167]	D3SENet Hybrid Network	Impressive 98.48% accuracy on three-class classification
[168]	SVM vs. CNN for COVID-19 Detection	Superior accuracy of 98.3% using end-to-end CNN model
[169]	COVID-19 mRNA Vaccines Analysis	Comprehensive understanding of mRNA vaccine prospects
[170]	Long-Term Follow-Up in COVID-19	Highlighted need for post-acute COVID-19 syndrome study
[171]	Efficacy of COVID-19 Boosters	Clarified importance of booster doses against Omicron
[172]	Deep TL-Based CT Image Analysis	Enhanced COVID-19 screening using chest CT images
[173]	Privacy-Preserved Medical Systems	Improved data transmission security with DL techniques
[174]	Ensemble Method for COVID-19 Detection	Robust detection outcomes with fine-tuned networks
[175]	ET-NET Ensemble Approach	Enhanced supervised classification for COVID-19
[176]	TL-Based ResNet Models for COVID-19	Improved detection using CT/CXR and TL techniques
[177]	Weights-Only TL for COVID-19 Detection	Reduced resource requirements for point-of-care devices
[178]	Transfer Learning for COVID-19	Achieved binary classification for Covid and Normal categories

Table 9 compiles a collection of publicly available datasets and their respective sources, serving as valuable resources for upcoming researchers and practitioners in the field. These datasets hold significant potential for advancing academic research and practical applications within the domain of ML and DL.

Table 9 Publicly Available Datasets and Sources for ML and DL

Dataset	Disease	URL
WDBC dataset	Breast cancer	https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic)
Pima diabetic dataset	Diabetes	https://www.kaggle.com/uciml/pima-indians-diabetesdatabase
Chronic kidney disease dataset	Kidney disease	https://archive.ics.uci.edu/ml/datasets/Chronic_Kidney_Disease
CT kidney dataset	Kidney disease	https://www.kaggle.com/datasets/nazmul0087/ct-kidney-dataset-normal-cyst-tumor-and-stone
Kidney stone detection	Kidney stone	https://github.com/yildirimoza/Kidney_stone_detection
COVID-19 radiography database	COVID-19	https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database
Tuberculosis (TB) chest w-ray	Tuberculosis	https://www.kaggle.com/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset/discussion/292712
Pneumonia chest x-rays	Pneumonia	https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia
BreakHis	Breast cancer	https://web.inf.ufpr.br/vri/databases/breast-cancer-histopathological-database-breakhis/
Cleveland	Heart disease	https://archive.ics.uci.edu/ml/datasets/heart+disease
Diabetes dataset	Diabetes	https://data.mendeley.com/datasets/wj9rwkp9c2/1
Biglycan breast cancer dataset	Breast cancer	https://data.mendeley.com/datasets/mprscwxb7/3
Breast Ultra-sound Images Dataset	Breast cancer	https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset
LIDC-IDRI	Lung cancer	https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=1966254
Mammographic image analysis society	Breast cancer	https://www.repository.cam.ac.uk/items/b6a97f0c-3b9b-40ad-8f18-3d121eeef1459
Figshare	Brain cancer	https://figshare.com/articles/dataset/brain_tumor_dataset/1512427
International skin imaging collaboration	Skin cancer	https://www.isic-archive.com/
PH2	Skin cancer	https://www.fc.up.pt/addi/ph2%20database.html

6 Research Challenges and Future Perspectives

This review gives an in-depth focus on the application of ML in intelligent medical applications. ML algorithms are widely used for the diagnosis of different diseases, however, researchers still face several challenges. In this section, the research challenges and future prospects of ML in medical applications have been discussed.

6.1 Challenges Related to Data Availability

1. **Insufficient Labeled Data:** ML models rely on large amounts of labeled data for training and evaluation. However, in the context of medical diagnosis, obtaining labeled data can be challenging and time-consuming. This scarcity of labeled data hinders the model's ability to generalize well and may lead to poor performance and excessive drift.
2. **Malicious Attacks:** ML models are vulnerable to malicious attacks where false data is intentionally fed to the model to deceive it into making incorrect predictions. These attacks can compromise the integrity and reliability of the model's outputs, posing significant risks in medical diagnosis and treatment decision-making.
3. **Noisy Data:** Noisy data, which contains errors, outliers, or corrupted information, can have a detrimental effect on the performance of ML models. The presence of noisy data can introduce biases, distort patterns, and increase the learning time of the model, ultimately affecting its accuracy and robustness in medical diagnosis tasks.

6.2 Challenges Related to ML Algorithms for Disease Diagnosis

1. **Disease Misdiagnosis:** Some diseases may exhibit similar visual appearances or symptoms, leading to misdiagnosis when relying solely on ML algorithms. For instance, diseases like COVID-19 and pneumonia share certain visual similarities, which can cause the algorithm to classify all test images into a single class, compromising the accuracy of the diagnosis.
2. **Lack of Interpretability:** The black-box nature of certain ML models poses challenges in terms of interpreting the model's predictions. The inability to understand and explain the decision-making process of the model hinders its clinical utility and reduces the trust and confidence clinicians may have in the predictions.

3. **Test Set Misclassification:** ML models may encounter challenges in accurately classifying patients in the test set, leading to misclassifications with potentially severe consequences. For example, if a patient is actually infected with COVID-19, but the model misclassifies them as having pneumonia, it can result in inappropriate treatment and delay in appropriate care.

6.3 Future Perspectives

Based on the challenges discussed in this review, we present potential research directions that future researchers can explore to address these challenges and further enhance the application of ML in disease diagnosis:

Enhanced Data Collection and Labeling: To address the challenge of insufficient labeled data, future research could explore techniques for data augmentation and active learning. Data augmentation methods can synthetically increase the size of the labeled dataset, while active learning can intelligently select the most informative data points for labeling, maximizing the model's learning efficiency.

Robustness and Adversarial Defense: To mitigate the risk of malicious attacks, researchers can focus on developing robust ML models with adversarial defense mechanisms. Adversarial training and detection techniques can help identify and handle adversarial examples, making ML models more secure and trustworthy in medical settings.

Data Preprocessing and Noise Handling: Innovative data preprocessing techniques and noise handling strategies can help reduce the impact of noisy data on ML models. Outlier detection, data cleaning, and robust feature extraction methods can enhance the model's ability to extract meaningful patterns from noisy medical data.

Multi-Model Ensembles: Combining multiple ML models in an ensemble can address the issue of disease misdiagnosis by leveraging the strengths of different algorithms. Ensemble methods can enhance the model's overall accuracy and robustness, reducing the risk of misclassifications in complex medical scenarios.

Explainable AI in Medical Diagnosis: The development of explainable AI techniques is crucial for increasing the interpretability of ML models in medical applications. Future research can focus on creating ML models that provide clear and transparent explanations for their predictions, enhancing trust and facilitating informed decision-making by healthcare professionals.

Transfer Learning and Few-Shot Learning: Transfer learning and few-shot learning approaches can be explored to overcome data scarcity challenges. By leveraging pre-trained models on large datasets, ML models can be

fine-tuned on limited medical data, enabling them to achieve better performance even with limited labeled samples.

Clinical Validation and Real-World Testing: To ensure the reliability and safety of ML-based medical applications, rigorous clinical validation and real-world testing are imperative. Future research should emphasize the deployment of ML models in clinical settings, where their performance can be evaluated under real-world conditions with diverse patient populations.

Collaborative Research and Data Sharing: Collaboration between researchers, healthcare institutions, and data-sharing initiatives can significantly enhance the availability of medical data for ML research. Encouraging data sharing while maintaining patient privacy can foster advancements in ML-based medical diagnosis.

In conclusion, while ML algorithms have shown promising results in intelligent medical applications, several challenges still need to be addressed to unleash their full potential. By focusing on data availability, model robustness, interpretability, and clinical validation, researchers can pave the way for safer, more accurate, and reliable ML-based diagnostic systems. These future perspectives can drive progress in medical diagnostics, revolutionizing patient care and ultimately improving health outcomes.

7 Conclusion

In conclusion, this review has provided a comprehensive overview of the application of ML methods in medical diagnostics, focusing on the early detection of various diseases including cancer, COVID-19, diabetes, kidney, and heart disease. By examining recent developments and published papers between 2014 and 2021, we have highlighted the growing significance of ML technology in disease diagnosis. The review also included a literature analysis that assessed the subject area, year of publication, and journal distribution of publications in the medical field. The analysis revealed a significant growth in the utilization of ML in disease diagnosis since 2018, with notable contributions from journals such as Plos One, Springer, Elsevier, Nature Scientific Reports, and IEEE Access. Furthermore, the review has identified future research directions and trends to address the challenges and enhance the effectiveness of ML in medical diagnostics. These include strategies to tackle data scarcity, the use of ensemble learning techniques, exploration of fuzzy classifiers and feature selection methods, and the incorporation of explainable AI for better understanding and interpretation of model predictions. By providing this comprehensive overview and insights into future trends, this review serves as a valuable resource for researchers in the field, guiding them towards the most effective utilization of

ML methods in early disease detection. Continued advancements in ML technology and its application in medical diagnostics hold great promise for improving healthcare outcomes and patient care in the future.

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Declarations

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